

# Rebuilding Instagram

## A complete surface, layout & flow teardown

Every screen, every route, every upload path, every measurement — documented for a team building a photo & video social network from zero.

Includes a live verification pass — see §0.4. The running app was inspected in a signed-in session on 18 August 2026: full CSS token dump, live element measurement, and the web create flow walked to the upload dialog. **Seven widely-published "facts" turned out to be wrong** — including Instagram's button colour, which is no longer #0095F6. Everything still carries a confidence tag; treat UNVERIFIED values as hypotheses, not facts.

| Prepared for   | Date           | Scope                                    | Verified                  |
|----------------|----------------|------------------------------------------|---------------------------|
| thegreatest.ai | 18 August 2026 | iOS · Android · Desktop web · Mobile web | Live session, 18 Aug 2026 |

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# How to read this document

Instagram is a moving target. This section tells you exactly how much to trust each number that follows.

## 0.1 Confidence legend

Every non-obvious claim in this document carries one of four tags. They appear inline, next to the figure they qualify.

| TAG         | MEANING                                                                                                                                                                        | HOW TO TREAT IT                                                                                      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| VERIFIED    | Confirmed against a citable published source, or extracted from a shipped artifact (Instagram's own CSS variables, Meta's developer media guidelines, official announcements). | Build on it. Re-check only if you're shipping more than ~6 months from now.                          |
| CONTESTED   | Credible sources disagree, or the feature is mid-rollout / in A/B test. The document says which way the evidence leans.                                                        | Pick the leaning value, but design so the alternative isn't a rewrite. Listed again in the Appendix. |
| UNVERIFIED  | Reconstructed from product knowledge and community measurement, not confirmable from an accessible source today.                                                               | <b>Hypothesis, not fact.</b> Confirm in DevTools before a designer builds to it.                     |
| MOBILE ONLY | Surface qualifiers, used where the two platforms genuinely diverge.                                                                                                            | Scope your MVP accordingly — see Part 7.                                                             |
| WEB ONLY    |                                                                                                                                                                                |                                                                                                      |

## 0.2 What could not be verified, and why

Two hard walls were hit while assembling this:

- instagram.com is unreachable** from the authoring environment. Direct requests are refused at the network proxy; `robots.txt` disallows crawlers on all app routes; `web.archive.org` mirrors of the CSS bundle are likewise blocked. **No live DOM or computed-style inspection was possible.**
- No logged-in session was available.** The browser bridge to the requesting user's Chrome was not connected at author time, so the create flow was reconstructed from documented walkthroughs rather than clicked through directly.

**The single highest-value hour you can spend**

Open `instagram.com` in Chrome, press `⌘I`, and inspect computed styles on four elements: `<body>`, the left nav rail, one feed `<article>`, and one profile grid tile. That settles every **UNVERIFIED** measurement in Parts 2 and 3 in one sitting. Everything else in this document — flows, inventories, limits, codecs — is sourced and does not need that pass.

## 0.3 The seven things sources actively disagree on

Resolve these before a designer commits. Full detail in the Appendix.

| QUESTION                                  | THE DISAGREEMENT                                                                                                        | LEAN                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Where is the "+" Create button on mobile? | Top-left of Home/Reels/Profile (3 sources) vs top-right beside the notifications heart (2 sources)                      | Live A/B. Design the top bar so either slot works.                                     |
| Maximum Reel length                       | 20 min (Instagram's own feature page + in-app camera) vs 15 min (Graph API) vs 3 min (recommendation-eligibility cliff) | <b>All three are real, separate limits.</b> Spec them independently.                   |
| Hashtag cap per post                      | 5 (better sourced) vs 3 (observed in test cells)                                                                        | 5. Treat 3 as a test.                                                                  |
| Can you post Stories from the web?        | 3 sources say no, 1 says yes                                                                                            | <b>No.</b> Requires mobile emulation or Business Suite.                                |
| Can you post Reels from the web?          | 4 say yes, 2 say no                                                                                                     | <b>Yes</b> — but with no audio library and no custom cover upload.                     |
| Profile grid reordering                   | Shipping (May–Jun 2026) vs removed (2026)                                                                               | Assume <b>not available</b> . Partial rollout at best.                                 |
| Primary button colour direction           | Staying <code>#0095F6</code> vs migrating to black/white vs purple accent                                               | Assume <code>#0095F6</code> . The other two are tests, and they contradict each other. |

**One fact the internet routinely gets wrong — get this right or your designers build the wrong thing**

The **profile grid tile** is **3:4** (portrait). The **feed post** is **4:5**. These are different numbers for different surfaces. Instagram changed the grid from 1:1 square to 3:4 in early 2025; it center-crops every post to 3:4 in the grid regardless of what ratio it was uploaded at. Half the guides online say the grid is "4:5." It is not. (*Measured live: 215.8 × 287.7 px = exactly 0.75. Confirmed.*)

0.4 Live verification pass — what changed after inspecting the running app

**A logged-in session became available mid-authoring**

Instagram.com was inspected directly in a signed-in Chrome session on **18 August 2026**: full CSS custom-property dump, computed styles, live element measurement, and the web create flow walked as far as the upload dialog. **Everything in this section is measured, not sourced.** Twelve claims were upgraded from reconstruction to fact; **seven were wrong and are corrected below.** Parts 1–6 have been updated in place — this table exists so you can see what moved.

| CLAIM                          | WHAT RESEARCH SAID                                                                                 | WHAT THE LIVE APP ACTUALLY DOES                                                                                                                                                                                                                                                      |
|--------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary button colour          | #0095F6 Instagram blue; the reported black/white and purple migrations were "probably tests"       | <b>WRONG.</b> <code>rgb(74, 93, 249)</code> = <b>#4A5DF9</b> — an indigo / blue-violet. Shipped, live, in the tokens and rendered on every Follow button and CTA. <b>Instagram blue is gone from buttons.</b> The "purple accent" reports were right.                                |
| Web vs app navigation          | "Treat web and app navigation as diverged — no evidence the Oct 2025 reorder reached the web rail" | <b>WRONG.</b> The web rail <b>HAS</b> adopted it: Home → Reels → Messages → Search → Notifications. Web and app are aligned.                                                                                                                                                         |
| Search and Explore             | Two separate rail items (magnifier + compass)                                                      | <b>WRONG.</b> Merged into one item — labelled "Search", links to <code>/explore/</code> . There is no compass icon.                                                                                                                                                                  |
| Create position in rail        | "Near the bottom of the left sidebar"                                                              | <b>WRONG.</b> Position 7 of 9, mid-rail, directly below Notifications.                                                                                                                                                                                                               |
| Grid gutter                    | ~4px                                                                                               | <b>WRONG.</b> Exactly <b>1px</b> , both axes, on profile and Explore.                                                                                                                                                                                                                |
| Content container              | 935px max-width                                                                                    | <b>SUPERSEDED.</b> Profile header and grid are capped at <b>680px</b> (rendering 650.5px); the outer shell is <b>1457px</b> . 935px is legacy.                                                                                                                                       |
| Accepted upload formats        | JPG, PNG, HEIC/HEIF, BMP, non-animated GIF                                                         | <b>CORRECTED.</b> The file input's own accept list: <code>image/avif</code> , <code>image/jpeg</code> , <code>image/png</code> , <code>image/heic</code> , <code>image/heif</code> , <code>video/mp4</code> , <code>video/quicktime</code> . <b>AVIF is in. GIF and BMP are not.</b> |
| Grid tile 3:4                  | 3:4, centre-cropped                                                                                | <b>CONFIRMED</b> — 215.8 × 287.7 px, ratio 0.750                                                                                                                                                                                                                                     |
| Collapsed rail ~73px           | ~73px                                                                                              | <b>CONFIRMED</b> — 48px item at x=12 → ~72px footprint                                                                                                                                                                                                                               |
| Button radius 8px, type 14/600 | 8px buttons, 14px/600                                                                              | <b>CONFIRMED</b> exactly, on the live CTA                                                                                                                                                                                                                                            |
| No <nav> landmark              | Reported by an accessibility audit                                                                 | <b>CONFIRMED</b> — <code>document.querySelector('nav')</code> returns null                                                                                                                                                                                                           |
| Live video from web            | Contested — "2 say yes, 1 says no"                                                                 | <b>RESOLVED: yes.</b> The web Create menu reads <b>Post · Live video · Ad.</b>                                                                                                                                                                                                       |

**The headline: Instagram has quietly rebranded its interaction colour**

#0095F6 — the blue that has meant "Instagram" in every clone tutorial, every design kit and every brand-colour listicle for a decade — **is no longer the button colour.** It has been replaced by #4A5DF9 , a saturated indigo that also serves as the outgoing DM bubble colour. Every source consulted for this document still says #0095F6 . If you had built from research alone, your app would have shipped looking a year out of date.

This is the clearest possible argument for spending the DevTools hour described in §0.2 on anything you copy from anyone.

# Surface map — every screen that exists

Instagram is not one app. It is four products sharing an account system: a feed reader, a short-video player, a messenger, and a camera. This part inventories all of it.

## 1.1 Platform matrix

Before anything else: understand that Instagram deliberately runs a **capability hierarchy**, not feature parity. Mobile is the product. Web is a viewer with a light posting path. Mobile web is a funnel to the app.

| CAPABILITY                      | IOS / ANDROID                                    | DESKTOP WEB                          | MOBILE WEB                                               |
|---------------------------------|--------------------------------------------------|--------------------------------------|----------------------------------------------------------|
| Browse feed, Explore, Reels     | Full                                             | Full                                 | Full                                                     |
| Post photo / carousel           | Full                                             | Yes, up to 20 slides                 | Degraded                                                 |
| Post Reel                       | Full editor                                      | Upload only <span>CONTESTED</span>   | No                                                       |
| Post Story                      | Full camera + stickers                           | No <span>CONTESTED</span>            | Partial (this is why device-emulation workarounds exist) |
| Go Live                         | Yes (≥1,000 followers, public)                   | Account-gated <span>CONTESTED</span> | No                                                       |
| AR effects, filters at capture  | Yes (Meta-owned only)                            | No                                   | No                                                       |
| Audio / music library at upload | Yes                                              | No                                   | No                                                       |
| DMs — text                      | Full                                             | Full                                 | Degraded                                                 |
| DMs — voice / video call        | Yes                                              | Yes                                  | No                                                       |
| Shopping / product tagging      | Yes                                              | No                                   | No                                                       |
| Scheduling                      | Native (posts, carousels, Reels — never Stories) | No native                            | No                                                       |
| Insights / analytics            | Full                                             | Partial                              | No                                                       |

Design lesson worth stealing

Instagram never tried to make the web app equal. It made the web app good at the two things people actually do on a laptop — *browse* and *reply to DMs* — and pushed everything creative to the phone, where the camera is. If you are resource-constrained, copy that asymmetry rather than fighting it.

## 1.2 Mobile navigation — the 2026 arrangement

The bottom tab bar was reordered in a staged rollout announced October 2025 and completed around February–March 2026. This is the current arrangement: VERIFIED

TAB BAR — left to right

1  
Home

2  
Reels

3 · NEW  
DMs

4  
Search

5  
Profile

× The "+" Create button is NO LONGER in the tab bar — it moved to the top bar. This is the most-complained-about change in the redesign.

Mobile tab bar, 2026. Create was evicted from the thumb zone; DMs took the centre slot.

| # | TAB              | ICON                               | BEHAVIOUR & NOTES                                                                                                                                                                                 |
|---|------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Home             | House outline (filled when active) | Stories tray + feed. Tap again = scroll to top. Wordmark dropdown switches feed mode: <b>Following</b> · <b>Favorites</b> (algorithmic Home is the default and cannot be permanently overridden). |
| 2 | Reels            | Play triangle in rounded square    | Was position 3 in the 2023–25 layout. India and South Korea test <i>cold-starting the app directly into this tab</i> .                                                                            |
| 3 | DMs              | Paper plane + unread count badge   | <b>New tab.</b> Took the centre slot Create used to hold. The legacy swipe-left-to-DMs gesture now lands on Reels instead — the single biggest source of user complaint.                          |
| 4 | Search / Explore | Magnifier                          | Shifted right one slot.                                                                                                                                                                           |
| 5 | Profile          | The user's own avatar              | Long-press opens the account switcher.                                                                                                                                                            |

**Stated rationale** (Adam Mosseri): DMs, Reels and recommendations drive almost all of Instagram's growth; more than 50% of total time on Instagram is spent in Reels or DMs. The redesign is an explicit bet that *consumption and conversation* beat *creation* in the navigation hierarchy.

### Historical tab bar, for context

| ERA        | ARRANGEMENT                                    | WHAT IT TELLS YOU                            |
|------------|------------------------------------------------|----------------------------------------------|
| 2020       | Home · Explore · Reels · <b>Shop</b> · Profile | Commerce bet. Create moved to top-right.     |
| Feb 2023   | Home · Search · Reels · + · Profile            | Shop tab killed; Create returned to centre.  |
| 2023–2025  | Home · Search · Reels · + · Profile            | The arrangement most people still picture.   |
| Oct 2025 → | Home · Reels · <b>DMs</b> · Search · Profile   | Create evicted entirely. Messaging promoted. |

1.3 Gesture model

Gestures are load-bearing on mobile — several destinations have *no* button affordance at all.

| GESTURE                                      | RESULT                                                                                            |
|----------------------------------------------|---------------------------------------------------------------------------------------------------|
| Swipe left / right anywhere in a tab         | <b>Horizontal tab paging</b> through Home → Reels → DMs → Search → Profile. New in this redesign. |
| Swipe right on the Home feed                 | Opens the Stories / Reel camera. Preserved.                                                       |
| <del>Swipe left to DMs</del>                 | <b>Removed.</b> Now lands on Reels.                                                               |
| Double-tap post or Reel                      | Like, with heart-burst animation                                                                  |
| Reels: press-and-hold left or right edge     | 2× playback-speed scrub                                                                           |
| Reels: single tap                            | Pause (added Mar 2026, shows a mute glyph)                                                        |
| Stories: tap left third / right two-thirds   | Previous / next frame                                                                             |
| Stories: hold                                | Pause                                                                                             |
| Stories: swipe down / swipe up               | Dismiss / open reply or link sticker                                                              |
| Long-press Profile tab                       | Account switcher                                                                                  |
| Long-press carousel slide (after publishing) | Drag to reorder (Mar 2026)                                                                        |
| Long-press DM send button                    | Schedule message                                                                                  |
| Long-press a Note bubble                     | Custom bubble colour                                                                              |

If you copy one thing here, copy the redundancy — and if you copy the removal, expect the backlash

Every gesture above has a button equivalent *except* swipe-right-to-camera. That single exception is Instagram's most-used creation entry point and it is invisible. The removal of swipe-left-to-DMs, by contrast, broke years of muscle memory and generated sustained complaint through December 2025. **Gestures are cheap to add and extremely expensive to move.**

1.4 Complete mobile screen inventory

Grouped by the tab that owns them. This is the real answer to "how many pages" — **roughly 110 distinct destinations**, not counting settings leaves.

**Home branch** (~18 destinations)

Feed · Stories tray · **Story viewer** · Story reply / reaction sheet · Feed-mode switcher (Following / Favorites) · Post detail · **Comments sheet** (+ photo comments, edit comment, pin own comment) · Likes list · Share sheet (DM / Story / Repost / external) · Post "... " menu (Save, Edit, Edit thumbnail, Replace audio, Archive, Insights, Boost, Delete, Not interested, Report) · **Notifications / Activity** · "Your Algorithm" panel

**Reels branch** (~12)

Reels feed (For You) · **Friends tab** (public content friends liked / created / reposted / commented on) · Following · Discover (iOS test) · Latest / Most-viewed sort · **Audio page** (with "Use on Edits") · Remix entry · Template entry · "Your Algorithm" panel · **Watch History** (Nov 2025) · Reels comments · "Your Feeds" centre test exposing six feeds: Following, Friends, Latest, Saved, Favorites, Suggested

**DM branch** (~20 — the densest area of the app)

Inbox (Primary / General / Requests + custom chat folders) · **Notes row** · **Map** entry · **Instant**s (star icon) · Broadcast channel list · Broadcast composer · **Thread view** — reactions, replies, voice notes (≤5 min), vanish mode, themes, nicknames, polls & quizzes in groups, events, games, draw, translation, message scheduling, AI summaries · Audio call · Video call · New-chat composer · AI characters ("Instagram AIs", smiley icon)

**Search branch** (~9)

Search bar → result tabs **Top** · **Accounts** · **Audio** · **Tags** · **Places** (+ Reels tab in hashtag contexts) · Explore grid · Topic pills · Explore "Your Algorithm" panel

**Profile branch** (~30, including settings roots)

Grid (3:4 tiles) · **Highlights moved into the main grid** (Sep 2025) · Reels tab · Tagged tab · **Reposts tab** (Aug 2025) · Trials tab · Profile banners / pills · Profile card (QR) · Edit profile · **Professional Dashboard** → Insights (Overview / Engagement / Audience, Apr 2026) · Archive (Stories / Posts / Live — Live saved 30 days) · Saved + Collections · Close Friends list · "Secret Friends" list (test) · **Settings & Privacy** → Accounts Center, Your Activity, Notifications, Data usage and media quality, Content preferences, Hidden Words, Language and Translations, Subscriptions, Branded Content / Creator Marketplace, Supervision

**Create branch** (~20 — fully detailed in Part 5)

Creation picker (POST / STORY / REEL / LIVE) · Gallery multi-select · Crop · Filters · Edit tools · Lux · Share screen · Advanced settings · Reels camera · Reels editor · Cover picker · Audio picker · Voiceover · Stories camera (Normal / Create / Boomerang / Layout / Hands-free / Dual / AI Images) · Sticker tray · Text tool · Draw tool · Live setup · Broadcast channel creation

1.5 Complete web route table

Instagram's web app is a real SPA with real URLs — routes matter because they determine what is shareable, what is indexable, and what can be deep-linked.

| ROUTE                                      | RENDERS                                                                                                                                                                       | CONF. |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| /                                          | Home feed — story tray, single-column feed, right rail (own account chip + "Suggested for you"), feed-variant switcher                                                        | V     |
| /explore/                                  | Explore mosaic grid                                                                                                                                                           | V     |
| /explore/tags/:tag/                        | Hashtag page                                                                                                                                                                  | V     |
| /explore/locations/                        | Location directory by country / city                                                                                                                                          | V     |
| /explore/locations/:id/:slug/              | Location feed; paginates via <code>? max_id=</code>                                                                                                                           | V     |
| /explore/search/keyword/?q=                | Keyword search results                                                                                                                                                        | U     |
| /explore/people/                           | Suggested accounts                                                                                                                                                            | U     |
| /reels/                                    | Full-bleed vertical player, one reel per viewport, scroll/arrow paging                                                                                                        | U     |
| /reels/audio/:id/                          | Audio page — every reel using a track                                                                                                                                         | U     |
| /reel/:shortcode/                          | Single reel permalink                                                                                                                                                         | V     |
| /p/:shortcode/                             | Post permalink. <b>Opens as a modal over the current route when navigated internally; a full page on direct load.</b> <code>? img_index=N</code> deep-links a carousel slide. | V     |
| /tv/:shortcode/                            | Legacy IGTV — 301s to <code>/p/</code> or <code>/reel/</code>                                                                                                                 | U     |
| /:username/                                | Profile — header + tab bar (Posts / Reels / Tagged; Saved on own profile)                                                                                                     | V     |
| /:username/reels/ · /tagged/               | Profile sub-tabs                                                                                                                                                              | U     |
| /:username/saved/ · /saved/:collection_id/ | Saved & collections (own account only)                                                                                                                                        | U     |
| /:username/followers/ · /following/        | Real routes that render as modal dialogs over the profile                                                                                                                     | U     |
| /stories/:username/:media_id/              | Story viewer — full-screen dark overlay                                                                                                                                       | V     |
| /stories/highlights/:highlight_id/         | Highlight viewer                                                                                                                                                              | U     |
| /direct/inbox/                             | DM inbox — two-pane: thread list left, conversation right                                                                                                                     | V     |
| /direct/t/:thread_id/ · /new/ · /requests/ | Thread · composer · message requests                                                                                                                                          | U     |

| ROUTE                                                                                                                                      | RENDERS                                             | CONF.          |
|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|----------------|
| /accounts/edit/                                                                                                                            | Edit profile — settings shell with its own left nav | <span>V</span> |
| /accounts/password/change/ · /privacy_and_security/ · /two_factor_authentication/ · /blocked_accounts/ · /close_friends/ · /manage_access/ | Settings leaves                                     | <span>U</span> |
| /your_activity/interactions/likes · /comments · /photos_and_videos · /account_history                                                      | Your Activity hub                                   | <span>P</span> |
| /accounts/login/ · /emailsignup/ · /password/reset/                                                                                        | Auth — public and indexable                         | <span>V</span> |
| /accounts/remove/request/permanent/ · /temporary/                                                                                          | Delete / deactivate                                 | <span>U</span> |
| /session/login_activity/                                                                                                                   | Login activity                                      | <span>U</span> |
| /legal/terms/ · /legal/privacy/                                                                                                            | Legal                                               | <span>V</span> |

**Not routes — overlays with no URL change**

The **Create flow**, the **Notifications panel**, the **Search panel**, and "Switch appearance". Search and Notifications open as *slide-over panels from the left rail*, and opening either **forces the rail to collapse to icon-only**. That collapse-on-panel-open behaviour is one of the most distinctive patterns in the web app and is worth copying — it buys you ~170px of horizontal room without a layout jump.

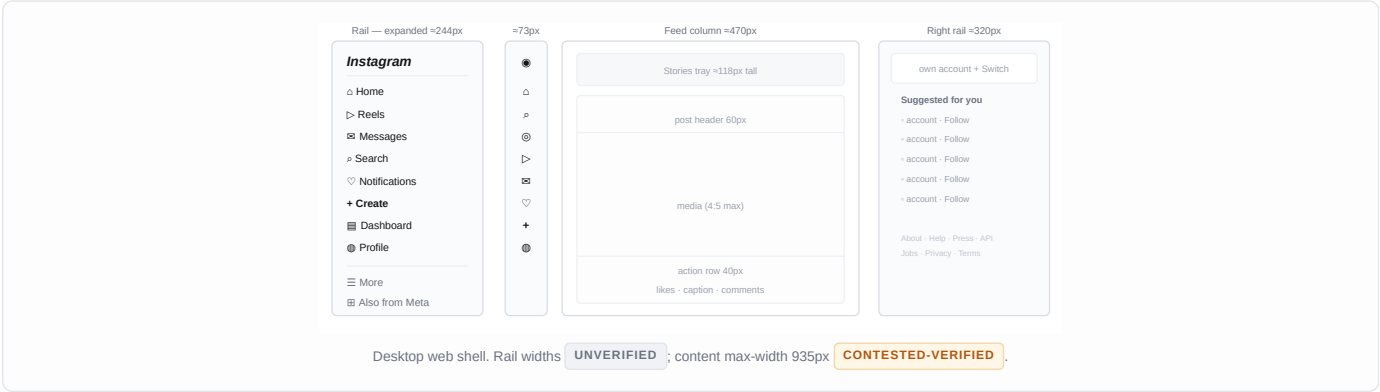
**Logged-out behaviour** VERIFIED

Worth specifying deliberately, because it is your entire SEO and viral-loop surface:

- Public profiles render bio, counts and roughly the **first 6–12 tiles** before a login wall
- Hashtag and location pages show roughly the **top 9**
- Stories, Highlights, Reels and all private accounts are **fully gated**
- Login interstitials are **dismissible with an X for a few interactions**, then become blocking — a deliberate soft-wall ramp, not a hard gate

**1.6 Desktop left rail**

Introduced 8–9 November 2022. Mosseri's stated goal was a layout built for large screens rather than a stretched mobile view. Home, Search, Explore, Messages and Notifications moved into a persistent left rail that **auto-collapses to icon-only based on available screen space**. VERIFIED



**Rail contents — measured live, 18 Aug 2026** VERIFIED

Read directly from the running app's DOM. Item hit area 48 × 48 px · icon 24 × 24 px · vertical pitch 56 px · collapsed rail footprint ≈72 px (48px item inset 12px from the edge).

| #                                              | ARIA-LABEL                                 | HREF           | NOTES                                                                                                                                      |
|------------------------------------------------|--------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1                                              | Instagram                                  | /              | Wordmark; collapses to the glyph mark                                                                                                      |
| 2                                              | Home                                       | /              | House icon, filled when active                                                                                                             |
| 3                                              | Reels                                      | /reels/        | Promoted to position 2 — the web rail has adopted the mobile reorder                                                                       |
| 4                                              | Messages                                   | /direct/inbox/ | Paper plane with an unread count badge                                                                                                     |
| 5                                              | Search                                     | /explore/      | Search and Explore are now ONE item. Labelled "Search", magnifier icon, but it navigates to /explore/. There is no separate compass entry. |
| 6                                              | Notifications                              | #              | Heart; opens a slide-over panel; forces rail collapse                                                                                      |
| 7                                              | New post (labelled "Create" when expanded) | #              | Mid-rail, directly below Notifications — not at the bottom. Opens a popover: Post · Live video · Ad                                        |
| 8                                              | Professional dashboard                     | #              | Professional accounts only                                                                                                                 |
| 9                                              | Profile                                    | /:user name/   | The user's avatar as the icon                                                                                                              |
| — pinned to the bottom, ~170px below Profile — |                                            |                |                                                                                                                                            |
| 10                                             | Settings (labelled "More")                 | #              | Hamburger glyph → Settings · Your activity · Saved · Switch appearance · Report a problem · Switch accounts · Log out                      |
| 11                                             | Also from Meta                             | #              | App-grid glyph. Not a Threads shortcut — a cross-app launcher.                                                                             |



#### Nine primary destinations, not eleven

Instagram's entire desktop navigation is nine items plus two utility entries, on a 56px pitch, in a 72px rail. It fits above the fold on any laptop without scrolling. The discipline is the design.

#### Do not copy Instagram's accessibility implementation

An independent audit found the rail is **not wrapped in a `<nav>` landmark**, uses `div[role=button]` instead of real `<a>` / `<button>` elements, has **no skip link**, story-tray arrows that are **not keyboard focusable**, malformed `role="menu"` / `role="menuitem"` relationships, invalid `aria-describedby` references, and insufficient contrast on the light-blue CTA against white. Semantic HTML costs you nothing extra and you will be held to a higher standard than an incumbent is.

#### Mobile web divergence

Mobile web is a genuinely separate layout, not the desktop rail at a small size:

- Top bar: wordmark left, DM icon right 
- **Bottom tab bar:** Home · Search · Reels · Notifications · Profile 
- Persistent "Open in app" interstitial and sticky banner
- The mobile-web bundle exposes a **Story composer the desktop bundle does not** — which is precisely why "post Stories from PC" workarounds all rely on device emulation

#### Corrected by live inspection: web and app are aligned, not diverged

Earlier research concluded the Oct 2025 reorder had not reached the web. **It has.** The desktop rail runs Home → Reels → Messages → Search → Notifications, matching the mobile tab bar's promotion of Reels and Messages. Instagram enforced parity on the *ordering* of shared destinations while keeping the *shell* different (rail vs tab bar) and the *capabilities* asymmetric. That is the right model to copy: consistent priorities, platform-native chrome.

# Layout system & measurements

The geometry that makes it feel like Instagram. Breakpoints are hard data; most widths are reconstructed — check the confidence tags.

## 2.1 Breakpoints — the strongest layout data available VERIFIED

A full CSS extraction of instagram.com found **135 unique media queries**. Three dominate by usage count:

| MEDIA QUERY                                | OCCURRENCES | WHAT IT GOVERNS                                                            |
|--------------------------------------------|-------------|----------------------------------------------------------------------------|
| (max-width: 767px)                         | 340         | <b>Mobile web.</b> Below this: no rail, bottom tab bar instead.            |
| (min-width: 768px) and (max-width: 1024px) | 319         | <b>Tablet.</b> Rail present but collapsed to icons; right rail suppressed. |
| (min-width: 1025px)                        | 321         | <b>Desktop.</b> Full three-column shell.                                   |
| 899px / 900px / 905px                      | —           | Secondary tier, likely governing right-rail appearance                     |

**Practical reading:** 767/768 is the mobile boundary; 1024/1025 is the tablet – desktop boundary. Two breakpoints carry almost the whole responsive system. You do not need five.

**CSS budget, for scale calibration**

Instagram.com ships **543 kB of CSS** — 27,992 lines, 10,314 rules, 16,235 declarations, **548 custom properties**, 539 unique colours, 78 font-family declarations, 140 @keyframes , and only 167 !important . That last number is the interesting one: a codebase that size with under 200 !important declarations is evidence of genuine atomic-CSS discipline. Class names are hashed ( x104yahw , \_aagv ) and keyframes auto-named — this is CSS-in-JS with a compile-time atomic extractor (StyleX).

## 2.2 Container widths

Only one width is confirmable from a shipped artifact: **max-width: 935px** on the primary content container, still present in a long-maintained community userstyle that tracks Instagram's DOM. 935px is Instagram's historic content width and appears to remain in force. CONTESTED-VERIFIED

| ELEMENT                         | VALUE          | CONF.          | NOTES                                                                                                                 |
|---------------------------------|----------------|----------------|-----------------------------------------------------------------------------------------------------------------------|
| Outer shell max-width           | 1457px         | <span>V</span> | <b>Measured live.</b> The app-level container cap.                                                                    |
| Profile header / grid max-width | 680px          | <span>V</span> | <b>Measured live</b> (renders at 650.5px). This supersedes the legacy 935px figure for profile and Explore.           |
| Legacy content container        | 935px          | <span>P</span> | Still present in the CSS; no longer governs the profile grid                                                          |
| Left rail item hit area         | 48 × 48px      | <span>V</span> | <b>Measured live</b>                                                                                                  |
| Left rail icon                  | 24 × 24px      | <span>V</span> | <b>Measured live</b>                                                                                                  |
| Left rail vertical pitch        | 56px           | <span>V</span> | <b>Measured live</b> — 48px item + 8px gap                                                                            |
| Left rail, collapsed            | ~72px          | <span>V</span> | <b>Measured live</b> — 48px item inset 12px. Active at ~768–1263px, and forced whenever Search/Notifications is open. |
| Left rail, expanded             | ~244px         | <span>U</span> | Hover/wide state, at ≥1264px viewport                                                                                 |
| Left rail, hidden               | <768px         | <span>V</span> | Bottom tab bar replaces it                                                                                            |
| Feed post card width            | ~470px         | <span>U</span> | Not measurable in the verification session — the test account followed zero accounts, so the feed rendered empty      |
| Right rail (suggestions)        | ~320px         | <span>U</span> | Suppressed below ~1025px                                                                                              |
| Create-post dialog              | 547 × 590px    | <span>V</span> | <b>Measured live</b>                                                                                                  |
| Post detail modal               | ~1035px × 90vh | <span>U</span> | Media pane sized to viewport height; comment pane ~405px fixed                                                        |

**Note how the profile grid is positioned**

At a 879px viewport the grid measured 650.5px wide starting at x=114.3 — exactly  $(879 - 650.5) / 2$  . **The grid is centred in the full viewport, and the collapsed rail floats over the left margin rather than pushing the content.** That is why the layout doesn't shift when the rail collapses on panel open. Small decision, large perceived-quality payoff.

Historic anchor (pre-Nov-2022): 935px container · 614px feed column · 293px right rail · 28px gutter. Useful only as a sanity check — every one of those numbers has since moved.

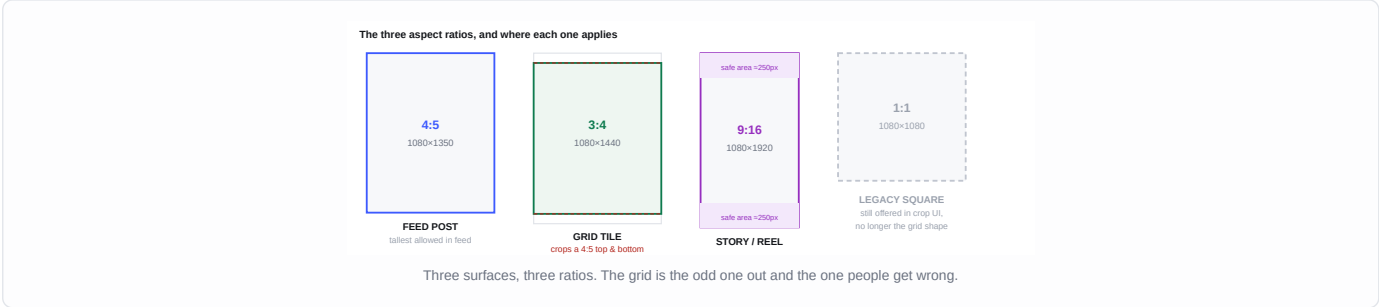
## 2.3 Profile grid VERIFIED

**The change was 1:1 → 3:4. Not 4:5.**

This is the most commonly garbled fact about modern Instagram and it will cost you a design cycle if you get it wrong.

| PROPERTY                | VALUE                  | NOTES                                                                                                                    |
|-------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Grid tile aspect ratio  | <b>3:4 portrait</b>    | Announced early 2025, fully rolled out through 2025                                                                      |
| Feed post aspect ratio  | <b>4:5</b> (1080×1350) | Different surface, different number                                                                                      |
| Grid crop behaviour     | Centre-crop, forced    | Every post is cropped to 3:4 in the grid <i>regardless of upload ratio</i> . A 4:5 upload loses a little top and bottom. |
| Crop safe zone          | 1012 × 1350 px         | Keep faces, text and logos inside this centred region of a 1080×1350 upload                                              |
| Grid-native upload size | 1080 × 1440 px         | Upload at this to control the grid crop exactly                                                                          |

| PROPERTY                | VALUE                       | NOTES                                                                                                                                                                                                           |
|-------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Columns                 | 3                           | Both mobile and web                                                                                                                                                                                             |
| Gutter                  | 1px                         | <span>MEASURED LIVE</span> — both axes, on profile and Explore. Not 4px. Instagram's grid is effectively hairline-separated, which is why it reads as a single photographic surface rather than a set of cards. |
| Measured tile size      | 215.8 × 287.7               | <span>MEASURED LIVE</span> at a 650.5px container — ratio 0.750 exactly                                                                                                                                         |
| Carousel thumbnail      | Slide 1, cropped to 3:4     |                                                                                                                                                                                                                 |
| Reel thumbnail          | Cover frame, cropped to 3:4 |                                                                                                                                                                                                                 |
| Story Highlight circles | 161 × 161 px                | Moved <i>into</i> the main grid, Sep 2025                                                                                                                                                                       |
| Grid reordering         | Assume unavailable          | <span>P</span> — one source logs it shipping Jun 2026, another says removed                                                                                                                                     |



2.4 Avatar size scale

Instagram stores **every** profile photo downsampled to 320 × 320 and renders it at these sizes. A tight, memorable scale — five sizes do the entire job. VERIFIED unless noted.

| CONTEXT                       | SIZE      | NOTES                                                                                |
|-------------------------------|-----------|--------------------------------------------------------------------------------------|
| Stored source resolution      | 320 × 320 | Everything downsamples to this                                                       |
| Profile page — desktop        | 150 × 150 | <span>P</span> — one source says 180; 150 is the long-standing value and more likely |
| Profile page — mobile         | 110 × 110 | Sources agree                                                                        |
| Story tray ring / avatar      | 56 × 56   | + ~3px gradient ring + 2px inner gap <span>U</span>                                  |
| DM thread list row            | 48 × 48   |                                                                                      |
| Right-rail suggestions        | ~44 × 44  | <span>U</span>                                                                       |
| Feed post header              | ~32 × 32  | + ~3px gradient ring when there's an unseen story <span>U</span>                     |
| Comment row                   | 32 × 32   |                                                                                      |
| Search results / Explore rows | 32 × 32   |                                                                                      |

**Story tray container:** ~118px tall, horizontal scroll, chevron arrows on desktop, username label ~11–12px truncated below each avatar. U

# Design tokens

Instagram's colour system is the most transferable thing in the entire product. The architecture matters more than the hex values.

## 3.1 Token architecture — copy this pattern regardless of your palette VERIFIED

Instagram.com ships two overlapping token layers:

**Layer 1 — --fds-\***

Meta's shared **Facebook Design System** primitives. Raw values, shared across Facebook, Instagram and Threads. Defined on `:root`.

Examples found in the bundle: `--fds-black: #000000`, `--fds-black-alpha-05 ...` `--fds-black-alpha-80`, `--fds-blue-05/30/40/60/70/80`, `--fds-button-text: #444950`, `--fds-comment-background: #f2f3f5`.

**Layer 2 — --ig-\***

Instagram's **semantic** layer on top. Names describe *role*, not colour: `--ig-primary-text`, `--ig-separator`, `--ig-error-or-destructive`.

**The clever bit:** values are stored as bare `R`, `G`, `B` triplets, not hex — so any consumer can composite alpha without a second token.

**The pattern, in code**

```
:root, __ig-light-mode {
  --ig-primary-text: 0, 0, 0;          /* bare RGB triplet, no rgb() wrapper */
  --ig-separator: 219, 219, 219;
}
__ig-dark-mode {
  --ig-primary-text: 222, 222, 222;    /* same names, different values */
  --ig-separator: 60, 60, 60;
}

/* consume at any opacity, from ONE token */
.caption { color: rgb(var(--ig-primary-text)); }
.caption-mute { color: rgba(var(--ig-primary-text), 0.6); }
.scrim { background: rgba(var(--ig-primary-text), 0.08); }
```

**Theme switching is class-based** — `__ig-light-mode` / `__ig-dark-mode` on a root element, not a media query. That means the user's in-app preference beats the OS preference, which is the correct hierarchy.

**Why the RGB-triplet trick is worth adopting**

Most design systems end up with `--text-primary`, `--text-primary-60`, `--text-primary-40`, `--text-primary-08` — four tokens for one colour, all of which must be maintained in both themes. Storing the triplet collapses that to one token per role and moves opacity to the call site, where it belongs. Instagram maintains 548 custom properties; without this trick it would be closer to 2,000.

## 3.2 The live token table — both themes MEASURED 18 AUG 2026

**This section was rewritten after live extraction. It supersedes every published Instagram colour reference.**

**97 --ig-\* / --web-\* / --post-\* tokens** dumped directly from the running app's stylesheets, with both the light ( `:root`, `__fb-light-mode` ) and dark ( `__fb-dark-mode` ) values side by side. Values are bare `R`, `G`, `B` triplets exactly as Instagram stores them.

**The single biggest finding in this document**

`--ig-primary-button` is **74, 93, 249** — **#4A5DF9**, a saturated indigo. It is **not** **#0095F6**. Instagram blue has been retired from buttons, links-as-CTAs, and the outgoing DM bubble. Every design kit, brand-colour site and clone tutorial still lists **#0095F6**. They are all out of date.



**--ig-primary-button · CURRENT**  
74, 93, 249 · #4A5DF9



**#0095F6 · SUPERSEDED**  
what every source still says

| TOKEN                      | LIGHT         | DARK          | ROLE                                                                                     |
|----------------------------|---------------|---------------|------------------------------------------------------------------------------------------|
| --ig-primary-background    | 255, 255, 255 | 12, 16, 20    | Page surface. Dark is <code>#0C1014</code> — a cool near-black, not <code>#0C0C0C</code> |
| --ig-secondary-background  | 243, 245, 247 | 37, 41, 46    | Recessed surface                                                                         |
| --ig-elevated-background   | 255, 255, 255 | 33, 35, 40    | Dialogs, sheets                                                                          |
| --ig-highlight-background  | 243, 245, 247 | 37, 41, 46    | Hover / pressed                                                                          |
| --ig-banner-background     | 255, 255, 255 | 38, 38, 38    | Banners                                                                                  |
| --ig-dropdown-background   | 243, 245, 247 | 37, 41, 46    | Dropdowns                                                                                |
| --ig-menu-background-color | 248, 249, 249 | 38, 43, 49    | Menus                                                                                    |
| --ig-primary-text          | 12, 16, 20    | 248, 249, 249 | <b>Neither pure black nor pure white.</b> <code>#0C1014</code> / <code>#F8F9F9</code>    |

| TOKEN                                           | LIGHT                              | DARK                       | ROLE                                                                      |
|-------------------------------------------------|------------------------------------|----------------------------|---------------------------------------------------------------------------|
| --ig-secondary-text                             | 106, 113, 122                      | 162, 170, 180              | #6A717A / #A2AAB4 — both carry a slight blue cast                         |
| --ig-tertiary-text                              | 199, 199, 199                      | 115, 115, 115              | Neutral grey, no cast                                                     |
| --ig-primary-icon                               | 12, 16, 20                         | 248, 249, 249              | Tracks primary-text                                                       |
| --ig-secondary-icon                             | 142, 142, 142                      | 142, 142, 142              | Theme-invariant                                                           |
| --ig-tertiary-icon                              | 199, 199, 199                      | 115, 115, 115              |                                                                           |
| --ig-primary-button                             | 74, 93, 249                        | 74, 93, 249                | #4A5DF9 — theme-invariant indigo CTA                                      |
| --ig-primary-button-hover                       | 65, 80, 247                        | 65, 80, 247                | #4150F7                                                                   |
| --ig-colors-button-primary-background--disabled | 184, 205, 255                      | 24, 32, 139                | Disabled CTA                                                              |
| --ig-secondary-button                           | 38, 38, 38                         | 245, 245, 245              | Secondary label                                                           |
| --ig-secondary-button-background                | 239, 239, 239                      | 85, 85, 85, .7             | Note the dark value carries alpha .7                                      |
| --ig-secondary-button-hover                     | 219, 219, 219                      | 54, 54, 54                 |                                                                           |
| --ig-tertiary-button-background                 | 255, 255, 255                      | 255, 255, 255              | Theme-invariant white pill                                                |
| --ig-tertiary-button-border                     | 219, 219, 219                      | 219, 219, 219              | Theme-invariant                                                           |
| --ig-colors-link-text                           | 65, 80, 247                        | 112, 141, 255              | #4150F7 / #708DFF — links also moved to indigo                            |
| --ig-link                                       | 0, 55, 107                         | 224, 241, 255              | Legacy deep-blue link token, still present                                |
| --ig-separator                                  | 219, 223, 228                      | 43, 48, 54                 | #DBDFE4 / #2B3036 — global borders                                        |
| --ig-elevated-separator                         | 219, 223, 228                      | 43, 48, 54                 | Navbar border                                                             |
| --ig-stroke                                     | 219, 219, 219                      | 54, 54, 54                 | Card borders                                                              |
| --ig-stroke-on-media                            | 255, 255, 255                      | 255, 255, 255              | Strokes over photos                                                       |
| --ig-focus-stroke                               | 168, 168, 168                      | 85, 85, 85                 | Focus ring                                                                |
| --ig-tablist-border · --ig-tab-underline-active | 219, 223, 228 · 12, 16, 20         | 43, 48, 54 · 248, 249, 249 | Profile tab bar                                                           |
| --ig-badge                                      | 255, 48, 64                        | 255, 48, 64                | #FF3040 — theme-invariant notification badge                              |
| --ig-error-or-destructive                       | 237, 73, 86                        | 237, 73, 86                | #ED4956 — errors, destructive rows, the liked heart                       |
| --ig-success                                    | 88, 195, 34                        | 88, 195, 34                | #58C322                                                                   |
| --ig-live-badge                                 | 255, 1, 105                        | 255, 1, 105                | #FF0169 — the LIVE pill. A distinct magenta, not the error red.           |
| --ig-close-friends-refreshed                    | 28, 209, 79                        | 28, 209, 79                | #1CD14F — the Close Friends green ring                                    |
| --ig-subscribers-only                           | 118, 56, 250                       | 118, 56, 250               | #7638FA — paid-subscriber purple                                          |
| --ig-facebook-blue                              | 53, 121, 234                       | 53, 121, 234               | #3579EA                                                                   |
| --ig-outgoing-message-bubble                    | 74, 93, 249                        | 74, 93, 249                | Same indigo as the primary button — one accent, used everywhere           |
| --ig-incoming-message-bubble                    | 243, 245, 247                      | 37, 41, 46                 | Tracks secondary-background                                               |
| --ig-hover-overlay                              | 12, 16, 20, .05                    | 255, 255, 255, .1          | Inverts across themes — dark ink at 5% on light, white at 10% on dark     |
| --ig-background-always-dark-blur                | rgba(43, 48, 54, .5)               |                            | Glass/blur surfaces over media                                            |
| --ig-toast-prism-background-color               | rgba(37, 41, 46, .95)              |                            | Toast fill — theme-invariant, near-opaque                                 |
| --ig-strong-shadow                              | 0 4px 32px 0 rgba(11, 16, 20, .16) |                            | One shadow token for the whole product. Large blur, low alpha, no spread. |
| --ig-input-border-radius                        | 12px                               |                            | Inputs                                                                    |
| --ig-input-border-width                         | 1px                                |                            |                                                                           |
| --ig-input-border-primary                       | 33, 35, 40                         | 248, 249, 249              | Focused input border                                                      |
| --ig-input-border-secondary                     | 219, 223, 228                      | 94, 100, 109               | Resting input border                                                      |
| --ig-input-text-and-icon-primary                | 12, 16, 20                         | 248, 249, 249              |                                                                           |
| --ig-input-text-and-icon-secondary              | 106, 113, 122                      | 162, 170, 180              | Placeholder                                                               |
| --ig-text-input-padding-vertical                | 20px                               |                            |                                                                           |
| --ig-toggle-background-on-prism                 | 12, 16, 20                         | 248, 249, 249              | Toggles are ink-coloured, not accent-coloured                             |
| --ig-toggle-background-off-prism                | 106, 113, 122                      | 162, 170, 180              |                                                                           |
| --ig-text-on-color · --ig-text-on-media         | 255, 255, 255                      |                            | Theme-invariant                                                           |
| --ig-temporary-highlight                        | 245, 251, 255                      | 0, 149, 246                | The only surviving use of #0095F6 in the token set                        |
| --post-separator                                | 239, 239, 239                      | 38, 38, 38                 | Between feed posts                                                        |
| --web-wash                                      | #FAFAFA                            | #18191A                    | Stored as hex, not a triplet — a legacy token                             |
| --web-always-black · --web-always-white         | 12, 16, 20 · 255, 255, 255         | same                       | Theme-invariant anchors. Note "always black" is #0C1014, not #000.        |
| --web-secondary-action                          | 224, 241, 255                      |                            | #E0F1FF                                                                   |

primary-background (dark)  
12, 16, 20 · #0C1014

secondary-background (dark)  
37, 41, 46 · #25292E

primary-text (dark)  
248, 249, 249 · #F8F9F9

secondary-text (dark)  
162, 170, 180 · #A2AAB4

secondary-text (light)  
106, 113, 122 · #6A717A

separator (light)  
219, 223, 228 · #DBDFE4

separator (dark)  
43, 48, 54 · #2B3036

live-badge  
255, 1, 105 · #FF0169

close-friends  
28, 289, 79 ·  
#1CD14F

subscribers-only  
118, 56, 250 ·  
#7638FA

error / liked heart  
237, 73, 86 ·  
#ED4956

badge  
255, 48, 64 ·  
#FF3040

Three structural observations worth more than the hex values

1. **Everything is cool-shifted.** "Black" is #0C1014, "white" text is #F8F9F9, greys carry a blue cast. Nothing is neutral. That is what makes the surface feel like glass rather than paper.
2. **Semantic colours are theme-invariant.** Badge, error, success, live, close-friends and subscribers hold identical values in both themes — only *surfaces* and *text* flip. That halves the maintenance burden and guarantees a red always means the same thing.
3. **One shadow, one accent.** A single --ig-strong-shadow and a single indigo doing CTA, link, and DM bubble duty. Compare that to the token sprawl in most design systems.

3.2b The old dark-mode set, for reference

The previously published values, now superseded by §3.2. Retained so you can recognise stale references when you meet them.

--ig-primary-background  
24, 24, 24 · #181818

--ig-secondary-background  
12, 12, 12 · #0C0C0C

--ig-elevated-background  
50, 50, 50 · #323232

--ig-highlight-background  
80, 80, 80 · #505050

--ig-primary-text  
222, 222, 222 · #DEDEDE

--ig-secondary-text  
180, 180, 180 · #B4B4B4

--ig-primary-button / --ig-link  
0, 149, 246 · #0095F6

--ig-separator  
60, 60, 60 · #3C3C3C

--ig-badge  
255, 48, 65 · #FF3041

--ig-error-or-destructive  
237, 73, 86 · #ED4956

--ig-success  
88, 195, 34 · #58C322

--ig-facebook-blue  
53, 121, 234 · #3579EA

| TOKEN                                   | RGB                     | HEX     | ROLE (ANNOTATED BY COMMUNITY REVERSE-ENGINEERING)   |
|-----------------------------------------|-------------------------|---------|-----------------------------------------------------|
| --ig-primary-background                 | 24, 24, 24              | #181818 | Alt / card background                               |
| --ig-secondary-background               | 12, 12, 12              | #0C0C0C | Main page background                                |
| --ig-elevated-background                | 50, 50, 50              | #323232 | Dialogs, sheets, elevated surfaces                  |
| --ig-banner-background                  | 50, 50, 50              | #323232 | Banners                                             |
| --ig-highlight-background               | 80, 80, 80              | #505050 | Hover / pressed states                              |
| --ig-temporary-highlight                | 245, 251, 255           | #F5FBFF | Transient highlight flash                           |
| --ig-primary-text                       | 222, 222, 222           | #DEDEDE | Main text — note: <i>not</i> pure white             |
| --ig-secondary-text                     | 180, 180, 180           | #B4B4B4 | Dimmed / meta text                                  |
| --ig-tertiary-text                      | 199, 199, 199           | #C7C7C7 | Tertiary                                            |
| --ig-text-on-color                      | 255, 255, 255           | #FFFFFF | Text on a filled button                             |
| --ig-text-on-media                      | —                       | —       | Text over photos/video (exists; value not captured) |
| --ig-link                               | 0, 149, 246             | #0095F6 | Links, @mentions, #hashtags                         |
| --ig-primary-button                     | 0, 149, 246             | #0095F6 | Follow / Share / Post CTA                           |
| --ig-secondary-button                   | 250, 250, 250           | #FAFAFA | Profile action links                                |
| --ig-separator                          | 60, 60, 60              | #3C3C3C | Global borders                                      |
| --ig-elevated-separator                 | 50, 50, 50              | #323232 | Navbar border                                       |
| --ig-stroke                             | 219, 219, 219           | #DBDBDB | Card borders                                        |
| --ig-focus-stroke                       | 168, 168, 168           | #ABA8A8 | Focus ring                                          |
| --ig-badge                              | 255, 48, 65             | #FF3041 | Notification count badge                            |
| --ig-error-or-destructive               | 237, 73, 86             | #ED4956 | Errors, destructive rows, <b>the liked heart</b>    |
| --ig-success                            | 88, 195, 34             | #58C322 | Success                                             |
| --ig-facebook-blue                      | 53, 121, 234            | #3579EA | Cross-app Facebook affordances                      |
| --post-separator                        | 24, 24, 24              | #181818 | Between feed posts                                  |
| --post-step-indicator                   | 168, 168, 168           | #ABA8A8 | Carousel dots (inactive)                            |
| --web-always-black / --web-always-white | 0, 0, 0 / 255, 255, 255 | —       | Theme-invariant anchors                             |
| --web-secondary-action                  | 224, 241, 255           | #E0F1FF | Soft blue action background                         |

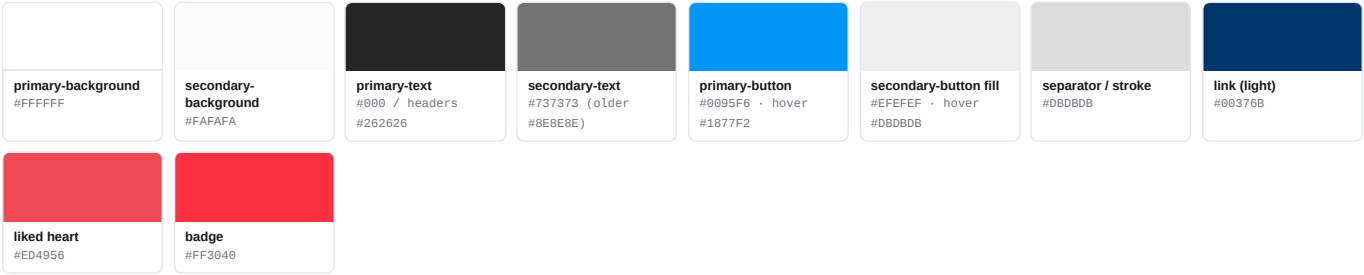
Notice what dark mode is *not*

Background is #0C0C0C, not #000000. Text is #DEDEDE, not #FFFFFF. Both edges are pulled in deliberately — pure black on pure white maximises contrast but produces halation and eye strain on OLED at night. Instagram gives up a few contrast points on purpose. Copy that restraint.

3.3 The old light-mode reconstruction SUPERSEDED BY §3.2

**Kept only as a cautionary exhibit**

These are the widely-circulated "Instagram light mode" hexes that every design resource publishes. **Live extraction showed several are wrong** — the primary button is indigo not blue, primary text is `#0C1014` not `#000000`, secondary text is `#6A717A` not `#737373`, and the separator is `#DBDFE4` not `#DBDBDB`. Use §3.2. This block shows you what the internet will tell you if you ask it.



**Independent corroboration:** a reverse-engineering of **Threads** — same Meta design system, same team, same era — documents `#0095f6` explicitly as "Instagram blue (legacy Meta variable)", alongside `#ffffff` canvas, `#fafafa` page wash, `#efefef` surface wash, `#000000` ink, `#65676b` ink-mute, `#999999` ink-soft, `#d5d5d5` hairline, `#31a24c` success, `#1877f2` Meta blue, `#fa383e` liked-cherry, `rgba(0,0,0,0.08)` shadow, and `#d24294` Meta focus pink — used *only* as the focus indicator ring, never as a CTA fill.

VERIFIED FOR THREADS

**Primary button colour is in flux — do not hard-code it everywhere**

One changelog logs Instagram "migrating to a black and white colour scheme for buttons" (April 2026). Another reports a shift toward **purple** accents. These contradict, so both are probably limited tests — but it means `#0095F6` is no longer a safe permanent assumption. Route every CTA through one semantic token so a brand change is a one-line diff.

3.4 The brand gradient VERIFIED

Meta's official brand page describes the gradient as built "from our vibrant 5-colour palette," made "from light," and confirms it is the source for the **story ring**, **app icon** and **brand communications** — but publishes **no hex values publicly**. The Meta Brand Resource Center gates the logo pack behind guideline acceptance and likewise publishes no hexes.



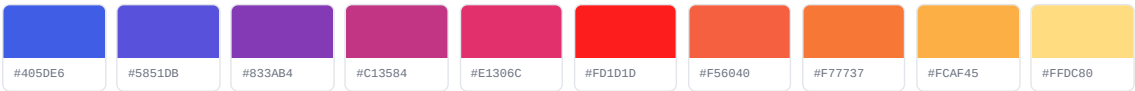
The 5-stop gradient in universal circulation — this is what story rings use in practice.

```
/* Story ring - 5 stops */
background: linear-gradient(45deg,
  #FEDA75 0%, /* yellow 254,218,117 */
  #FA7E1E 25%, /* orange 250,126, 30 */
  #D62976 50%, /* magenta 214, 41,118 */
  #962FBF 75%, /* purple 150, 47,191 */
  #4F5BD5 100% /* blue 79, 91,213 */
);
```

**3-stop simplification** used on Meta's own marketing surfaces:



The 10-colour legacy brand spectrum (unchanged since 2016)



Royal Blue · Blue · Purple · Dark Pink · Rose · Red · Dark Orange · Orange · Yellow · Light Yellow. Sources disagree marginally on two entries: `#5851DB` vs `#5B51D8`, and `#F77737` vs `#FD7720`. The first of each is more commonly cited.

**Do not ship this gradient in your app**

This exact five-stop warm-to-cool sweep is Instagram's registered trade dress. Reproducing it on a competing social product is the single most legally exposed thing you could copy from this document — it is more identifying than the wordmark. Build a gradient with a genuinely different hue path (e.g. cool-to-warm, or a two-hue analogous ramp). See Part 8.

3.5 Typography

Web font stack READ FROM COMPUTED STYLE, 18 AUG 2026

```
/* computed on document.body */
font-family: -apple-system, "system-ui", "Segoe UI", Roboto,
             Helvetica, Arial, sans-serif;
font-size:   14px;           /* body base */

/* root classes observed */
html class="_9dls _ar44 js-focus-visible _aa4d __fb-dark-mode"
body class="_ar45 system-fonts--body sf"
```

Note the body class literally reads `system-fonts--body`. The theme class is `__fb-dark-mode` — **a Facebook-namespaced class, not an Instagram one**, confirming the shared Meta design system runs the theming. Instagram's CSS also carries a documented `'Fix for Mac Chrome 80'` shim family among its 78 font-family declarations.

**Base body size is 14px** — measured, not inferred. Native app UI is SF Pro on iOS and Roboto on Android. **Instagram uses the system font everywhere in product UI.** No webfont, no FOIT, no loading cost, no licence.

Motion curves EXTRACTED LIVE

The shared `--fds-animation-*` easing set, identical in both themes — six curves for the entire product:

| TOKEN                                     | CUBIC BÉZIER                 |
|-------------------------------------------|------------------------------|
| --fds-animation-enter-exit-in             | cubic-bezier(.14, 1, .34, 1) |
| --fds-animation-enter-exit-out            | cubic-bezier(.45, .1, .2, 1) |
| --fds-animation-expand-collapse-in / -out | cubic-bezier(.17, .17, 0, 1) |
| --fds-animation-move-in / -out            | cubic-bezier(.17, .17, 0, 1) |
| --fds-animation-passive-move-in / -out    | cubic-bezier(.5, 0, .1, 1)   |
| --fds-animation-fade-in / -out            | cubic-bezier(0, 0, 1, 1)     |

Two things to steal: **fares are linear** (`0, 0, 1, 1`) — easing an opacity change reads as sluggish; and **entrances and exits use different curves** (`.14, 1, .34, 1` in, `.45, .1, .2, 1` out) — things arrive with overshoot-free deceleration and leave with acceleration. Instagram also ships a duration scale (`--ig-duration-extra-extra-short-in: .1s` and siblings) rather than hard-coded milliseconds.

**Instagram Sans is not available to you**

The proprietary geometric sans launched May 2022 (squirrel-derived letterforms, Light/Regular/Medium/Bold + Condensed, Latin/Arabic/Japanese) is **brand-and-marketing only, not licensable, and not used for body text in the app**. It appears in Stories text styles and brand comms. Pick a geometric alternative for your own brand layer — and keep system fonts for product UI, exactly as Instagram does.

Type scale UNVERIFIED RECONSTRUCTION

Threads — same design language, same team — uses a documented compact scale: **20/700 page heading, 15/400 body, 15/600 emphasis, 12/400 caption**. Instagram web sits about one notch smaller. Reconstruction:

| ELEMENT                          | SIZE / WEIGHT              | COLOUR TOKEN   |
|----------------------------------|----------------------------|----------------|
| Post header username             | 14px / 600                 | primary-text   |
| Post caption & comment body      | 14px / 400                 | primary-text   |
| Caption username prefix          | 14px / 600                 | primary-text   |
| Likes count                      | 14px / 600                 | primary-text   |
| Timestamp, "View all N comments" | 12px / 400                 | secondary-text |
| Sidebar nav label                | 16px / 400 (600 active)    | primary-text   |
| Profile stat numbers             | 16px / 600                 | primary-text   |
| Profile username (header)        | 20px / 400                 | primary-text   |
| Story tray label                 | 12px / 400                 | secondary-text |
| Buttons                          | 14px / 600                 | text-on-color  |
| Base line-height                 | 18px body · 1.5 captions — |                |

Note how few sizes that is. Effectively 12 / 14 / 16 / 20 px and two weights (400 / 600). A four-step scale carries the entire product. Resist the urge to add more.

Radii, elevation, motion UNVERIFIED

| PROPERTY      | VALUES                                                                                                                                                                                                                                                                   |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Border radius | 4px inputs & small controls · <b>8px buttons and post cards</b> · 12px modals & sheets · 20px pills (search field, some Follow pills) · 50% avatars. <i>Threads uses a 1000px full-pill convention plus 18px cards — Instagram appears to be drifting pill-ward too.</i> |
| Elevation     | Feed cards on web have <b>no drop shadow</b> — separation is a 1px <code>--ig-separator</code> rule only. Shadow appears on dialogs and popovers: <code>rgba(0, 0, 0, 0.08)</code> family.                                                                               |
| Scrim         | Modal backdrop <code>rgba(0, 0, 0, 0.65)</code>                                                                                                                                                                                                                          |
| Motion        | Heart-burst on double-tap (scale + fade, ~600ms) · slide-over panel translateX · story progress-bar linear fill · carousel snap. Durations short: 150–250ms for UI, longer only for celebratory moments.                                                                 |



#### The most transferable takeaway in Part 3

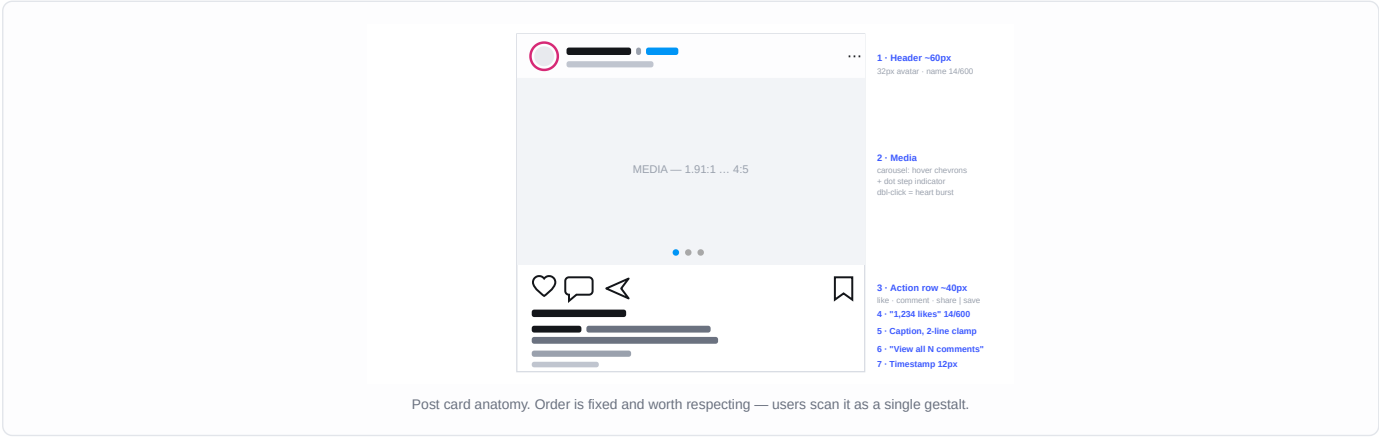
Two-layer tokens ( `--fds-*` primitives → `--ig-*` semantic aliases), stored as **R, G, B triplets** so alpha composites at the call site, with **class-based** theme switching. That architecture is genuinely good design-systems work and costs you nothing to adopt. The hex values are the least valuable part of this section.

# Component library

Roughly fifteen components carry the entire product. Build these well and the rest is composition.

## 4.1 The post card — the most important component you will build

Vertical stack, ~470px wide on desktop, full-bleed on mobile, separated from its neighbours by a 1px rule. No shadow, no card border on web.



| # | ZONE                        | CONTENTS                                                                                                                                                                                                                                                            |
|---|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Header ~60px                | 32px avatar (gradient ring if unseen story) · username 14/600 linking to profile · optional "• Follow" inline link · verified badge · "• 2d" relative timestamp · location line beneath username when present · right-aligned "... " opening the post options sheet |
| 2 | Media                       | Image / video / carousel. Carousel adds hover chevrons and a dot step indicator (inactive #A8A8A8 , active #0095F6 ). Double-click fires a heart-burst. Video shows a mute toggle; <b>desktop gained a media volume control April 2026</b> .                        |
| 3 | Action row ~40px            | Left group: heart (Like) · speech bubble (Comment) · paper plane (Share). Right-aligned: bookmark (Save). Liked state = filled heart in #ED4956 .                                                                                                                   |
| 4 | Likes                       | "Liked by X and N others" / "N likes", 14/600, opens a likes dialog                                                                                                                                                                                                 |
| 5 | Caption                     | username 14/600 + text 14/400, clamped to ~2 lines with a "more" toggle. @mentions and #hashtags in --ig-link .                                                                                                                                                     |
| 6 | Comments                    | "View all N comments" link, then 1–2 preview comments                                                                                                                                                                                                               |
| 7 | Add-comment row + timestamp | Emoji button, "Add a comment..." input, "Post" text button (dim until non-empty). Timestamp 12px secondary.                                                                                                                                                         |

## 4.2 Post detail modal

Opened by internal navigation to /p/:shortcode/ — the URL changes but the underlying route stays mounted behind a scrim. That single behaviour is what makes Instagram feel fast: you never lose your feed scroll position.

- Scrim rgba(0,0,0,0.65) ; modal ~1035px × 90vh
- **Media pane left** — sized to viewport height, black letterbox background
- **Thread pane right** — fixed ~405px: sticky header (avatar, username, Follow, ...), scrollable comment list, pinned footer with action row, likes, timestamp and the add-comment input
- Collapses to a single stacked column on narrow viewports
- Close = X top-right of viewport, Esc , or scrim click. Left/right chevrons outside the modal page between posts.

## 4.3 Comment thread

- **Row:** 32px avatar · body ( username 14/600 inline + text 14/400) · meta line 12px secondary (relative time · "N likes" · "Reply" · "... " for own comments · translate) · right-aligned 12px heart
- **Replies** collapse under – View replies (N) with a leading dash rule, indented ~44px; expanding paginates in batches
- Input supports emoji picker and @-mention autocomplete; "Post" activates on non-empty
- **Pinned comments** render first with a pin glyph — the creator's most-used moderation tool
- Photo comments and comment editing both exist on mobile

## 4.4 Story tray & viewer

- Tray**
- ~118px tall, horizontal scroll, 56px avatars with a ~3px gradient ring and a 2px background-coloured inner gap. Username label ~11–12px truncated below. Desktop adds chevron arrows (which are, as noted, not keyboard focusable — fix that in yours).
- Ring states you must design:** unseen (full gradient) · seen (grey #DBDBDB ring) · Close Friends (solid green) · live (red "LIVE" pill) · your own story (plus badge to add).
- Viewer**
- Full-viewport takeover, near-black background
  - Centre card is 9:16, 12px radius, capped by viewport height

- **Segmented progress bars** at the top — one segment per story, 2px tall, track at 30% opacity, fill at 100%
- Below the bars: avatar + username + "3h" + mute + ... + X
- **Desktop-only affordance:** adjacent users' stories render as scaled-down (~0.55) dimmed cards to left and right — a genuinely distinctive touch that makes the desktop viewer feel considered rather than ported
- Footer: "Send message" pill input + heart + share. Arrow keys page between users.

4.5 DM inbox & thread

| REGION             | CONTENTS                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Left column ~350px | Header: own username + dropdown chevron, "new message" pencil · horizontally scrolling <b>Notes</b> row of avatar bubbles · <b>Primary / General / Requests</b> or chat-folders segmented control (chat folders shipped Sep 2025) · thread rows ~56px tall: 48px avatar, name 14/600, last-message preview + ". 2h" 14/400 truncated, unread blue dot right-aligned                                                   |
| Right column       | Thread header with avatar, name, <b>audio-call and video-call buttons</b> · message list: outgoing right-aligned blue ( #3797F0 / #0095F6 family), incoming left-aligned - -ig-elevated-background grey, both pill-radius, grouped by sender with the avatar on the last bubble of a run · hover reveals emoji-react and reply/more · composer with emoji, text input, mic, image, sticker/GIF and a heart quick-send |
| Empty state        | Paper-plane illustration + "Your messages" + "Send a message to start a chat" + blue "Send message" button                                                                                                                                                                                                                                                                                                            |

2026 additions: Draw in DMs (Oct 2025) · DM scheduling and translation for business profiles (Jul 2025) · request filters (Apr 2026) · AI summaries · polls, quizzes, events and games in group threads · vanish mode · themes and nicknames · voice notes up to 5 min.

4.6 Dialogs, sheets, panels, toasts

**Centred options sheet** — IG's dominant pattern

--ig-elevated-background, 12px radius, **no header bar**. A vertical stack of full-width ~48px rows separated by 1px --ig-separator. Destructive row in --ig-error-or-destructive, bold. Last row is "Cancel" in plain weight.

Used for: post ..., profile ..., comment ..., share targets. Copy this wholesale — one component covers a dozen menus.

**Confirmation dialog**

~400px wide, centred icon, title 16/600, body 14/400 secondary, then the same stacked-row buttons. Consistency with the options sheet is deliberate.

**Slide-over panel** WEB

Search and Notifications. Enters from the left rail with a translateX transition, ~397px wide, own scroll container, dismissed by outside click. **Forces the rail into collapsed state while open.**

**Toast & bottom sheet**

**Toast:** bottom-centre on mobile, bottom-left on desktop. Dark #262626 pill, white 14/400, ~3s auto-dismiss, optional "Undo".

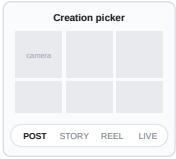
**Bottom sheet** MOBILE: 12px top radius, drag handle, backdrop scrim, snaps to content height.

# Create & upload — the complete flows

This is the part you asked for in detail. Every path from "I have a file" to "it's published," on every surface.

## 5.1 Entry points

Tapping + opens the **creation picker**: a gallery-first full-screen media grid with a horizontal mode selector along the bottom.



**Creation picker**

camera

POST STORY REEL LIVE

**Platform difference at the picker**

Android — shows a camera ICON you tap to shoot  
iOS — shows the LIVE camera as the first cell in the grid

**Also reachable from "+" in some builds**

Story Highlight · Broadcast channel · Add note

**Gates to design for**

LIVE — public account, ≥1,000 followers (since 1 Aug 2025)  
BROADCAST — Professional account; 10k follower gate contested

A new post composer began rolling out in August 2026

Mobile is mid-redesign on the composer itself. Everything below describes the flow as documented through mid-2026; expect visual churn on these screens specifically, and less on the underlying step sequence — the step sequence has been stable for years.

## 5.2 Photo post — step by step

- 1

**Gallery picker**

Full-screen media grid, most-recent first. Album selector at top. A "select multiple" toggle enters carousel mode. Camera entry differs by platform (see above).

Header: [Cancel] "New post" [Next →]
- 2

**Crop**

Pinch-to-zoom and drag to reposition. In-app crop presets: **1:1 square** · **4:5 portrait** · **1.91:1 landscape**, plus "Original". The feed itself accepts anything from 1.91:1 to 4:5; 9:16 portrait images are also accepted via the API.

Bottom-left: [expand icon] [multi-select icon] [camera icon]
- 3

**Filter**

Horizontal tray of live-preview thumbnails. Each filter applies at 100% by default; **tap the filter name a second time to reveal an intensity slider**. "Manage" at the end of the tray lets you drag to reorder filters and hide ones you never use via checkmarks — a small, excellent piece of personalisation almost nobody copies.
- 4

**Lux** MOBILE ONLY

Sun icon at the *top* of the edit screen, above the Filter/Edit tabs — deliberately separated from everything else. A single bipolar slider: right = more contrast and vibrancy with shadow/highlight recovery; left = faded and flat. It is Instagram's one-tap "make this look better" control and it predates every AI enhance feature.
- 5

**Edit — 13 tools**

Full catalogue in §5.10. Bipolar sliders are centre-detented; unipolar sliders are left-anchored.
- 6

**Share screen**

Caption, tagging, collaborators, location, music, audience, cross-post, advanced settings. Full field list in §5.8.

## 5.3 Carousel

| PROPERTY                         | VALUE                                                                                                                             | NOTES                                                                                                                         |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Max items — in app               | 20                                                                                                                                | Raised from 10                                                                                                                |
| Max items — via API / schedulers | 10                                                                                                                                | Third-party tools still cap here because Meta's API does. <b>Design for both.</b>                                             |
| Mixed media                      | Yes                                                                                                                               | Photos and videos freely mixed                                                                                                |
| Aspect ratio rule                | Slide 1 sets the ratio; every later slide is cropped to match. This applies on web too, and it is the number-one carousel gotcha. |                                                                                                                               |
| Video per slide                  | 60 s                                                                                                                              |                                                                                                                               |
| Per-item editing                 | Both modes                                                                                                                        | Apply one filter to everything, or edit each slide individually                                                               |
| Cover                            | Slide 1                                                                                                                           | No separate cover picker for photo carousels (Reels have one)                                                                 |
| Post-publish reorder             | Yes (2026)                                                                                                                        | Long-press-drag. You may <b>delete down to a minimum of 2 slides</b> , but you <b>cannot add a new slide or swap a file</b> . |
| Per-slide captions               | Yes (May–Jul 2026)                                                                                                                | Each slide can carry its own caption                                                                                          |

5.4 Video post vs Reel — an important structural fact

Since July 2022 there is no such thing as a "feed video"

All new video uploads to the feed become Reels. The only place a non-Reel video object still exists is *inside a carousel*. If you are modelling your data schema, do not create separate `Video` and `Reel` entities — Instagram collapsed them, and the collapse simplified their product enormously. One short-video object, surfaced in multiple feeds.

5.5 Reels camera & editor

Camera toolbar — the left rail

| TOOL         | BEHAVIOUR                                                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Audio        | Music library, original audio, saved audio, imported audio. "Saved" shortcut added iOS Aug 2026.                                            |
| Length       | Legacy presets <b>15 / 30 / 60 / 90 s</b> . <b>Camera recording ceiling raised to 20 minutes</b> Nov 2025.                                  |
| Speed        | 0.3x · 0.5x · 1x · 2x · 3x (some builds 4x)                                                                                                 |
| Layout       | Split-screen grids for multi-take composition                                                                                               |
| Timer        | Countdown before record + a duration stop-point for hands-free capture. Refreshed UI Nov 2025.                                              |
| Dual         | Simultaneous front + rear camera                                                                                                            |
| Effects      | AR filter tray — <b>Meta-owned effects only</b> (see the Meta Spark note below)                                                             |
| Align        | Ghosts the last frame of the previous clip so you can line up transition objects. The single most-loved tool in the set.                    |
| Green screen | Background replacement. Nov 2025 improved edge detection around hair and fine detail; real-time compositing.                                |
| Touch-Up     | Changed from a boolean toggle to a <b>slider</b> (Nov 2025). Skin-smoothing intensity, applied at capture and <b>baked in permanently</b> . |
| Undo         | Removes the most recent clip from the timeline; tap repeatedly. <b>Not reversible</b> .                                                     |
| Teleprompter | Scrolling script overlay while recording (iOS, Apr 2026)                                                                                    |
| Gallery      | Import existing clips                                                                                                                       |

Editor screen

Clip timeline with per-clip trim, reorder and add · **audio mixer** (original vs music volume, loudness matching) · **Voiceover** · **text with per-word / per-clip timing** (tap a text layer, then set start and end on the timeline) · stickers · auto-generated captions · filters · **Cover picker** (choose a frame or upload; editable post-publish since Mar 2026) · **Templates** (reuse another Reel's music and cut timing, drop in your own clips) · **Remix** (record alongside someone else's Reel) · Drafts.

Third-party AR effects no longer exist — plan accordingly

Every third-party AR effect was deleted on **14 January 2025** when Meta shut down Meta Spark (announced Aug 2024). Only Meta-owned first-party effects remain. Branded campaign filters, "which X are you" spinners, and the entire community beauty-filter ecosystem are gone. If you were planning a creator-effects platform as a differentiator, that is now an *open* space — Meta vacated it.

Edits — the companion app, and what it signals

Meta launched **Edits** (iOS + Android, free) on 22 April 2025: 10-min in-app capture, teleprompter, **keyframes** (position/rotation/scale), freeze frame, reverse, lip sync, beat markers, cut-silences, masks, SAM-based object segmentation with effect tracking, AI Restyle, colour wheels and curves, storyboards, project versions, **export up to 15 min watermark-free**, native Reels analytics, and draft sharing via DM. 130+ features shipped since launch; a desktop version is in development.

**The signal:** Instagram concluded the serious editing tools do not belong in the main app. If you are building, resist stuffing a professional NLE into your composer. Ship a good-enough in-app editor and let power users bring finished files.

5.6 Stories

Camera modes

**Normal (Standard)** · Create · Boomerang · Layout · Hands-free · Dual · **AI Images**. *Level* (horizon guide) and *Superzoom* appear in some builds. Layout offers **3, 4 or 6-photo** collage grids.

Sticker catalogue — the whole reason Stories works

| CATEGORY    | STICKERS                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interactive | <b>Poll</b> (2 options) · <b>Quiz</b> (multiple choice, one correct) · <b>Questions</b> (free text) · <b>Emoji Slider</b> · <b>Countdown</b> (to a date/time) · <b>Add Yours</b> (public prompt chain) · Add Yours Music · Crystal Ball (test)                                                                                                                                                                                         |
| Media       | <b>Music</b> (album art / lyrics / waveform display styles, tap to cycle) · GIF (GIPHY) · <b>Cutout</b> (auto subject extraction from your camera roll or eligible public posts, saved to a tray for reuse) · <b>Frames</b> (polaroid; shake-to-reveal, auto date stamp) · <b>Reveal</b> (blurred content unlocked by sending a DM) · Photostrip (2026) · Highlights sticker · Voice sticker · Before/After (AI edit comparison, 2026) |
| Metadata    | Location · Hashtag · @Mention · Time · Weather/Temperature                                                                                                                                                                                                                                                                                                                                                                             |
| Utility     | <b>Link</b> (URL — open to all accounts since 2023) · DM Me · <b>Captions</b> (auto-transcribe audio) · Product · Get Orders / Food Orders · Donation / Fundraiser · Notify / Reminder                                                                                                                                                                                                                                                 |
| Templates   | Add Yours → Template mode (Oct 2025)                                                                                                                                                                                                                                                                                                                                                                                                   |

Reveal, Frames, Cutouts and Add Yours Music all shipped on one day — 3 May 2024

Instagram ships stickers in batches, not one at a time. There is a lesson in that: interactive stickers are cheap to build individually and their value is combinatorial. A batch of four lands as "Stories got better." One sticker lands as nothing.

Text, draw, and duration

**Text styles:** Classic · Modern · Neon · Typewriter · Strong · Deco · Meme · Squeeze · Bubble · Poster · Calligraphic · Geometric, plus newer additions Rosalia, Comic, Retro Chrome, Ghost, seasonal/licensed styles, Korean fonts (Feb 2026), and custom bio/profile fonts. Controls: size slider, colour picker + eyedropper, alignment, background/highlight fill, opacity blending (Jun 2026).

**Draw tools:** pen, marker/highlighter, neon glow, eraser, arrow/chisel tip; colour picker + eyedropper; size slider. Also available inside DMs since Oct 2025.

| LIMIT               | VALUE  | NOTES                                                                                                                                          |
|---------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Photo frame display | ~5–7 s | Extends to match attached music (~15 s)                                                                                                        |
| Video per segment   | 60 s   | Raised from 15 s in 2022. Longer uploads <b>auto-split into consecutive 60-s segments</b> — a lovely bit of UX that removes a decision.        |
| Segments per 24 h   | 100    |                                                                                                                                                |
| Lifespan            | 24 h   | Then auto-archives. <b>48 h with Instagram Plus</b> ("Story Extend").                                                                          |
| Audio in a story    | 45 s   | Raised Oct 2025. A separate Apr 2026 note cites the <i>music sticker</i> clip at 20 s — different limits, both build-dependent. <span>P</span> |
| Draft expiry        | 7 days | Sep 2025                                                                                                                                       |

2025–26 Stories additions: public Story Comments (mutual followers, 24 h visibility) · screenshot prevention / view-once · **editing a story after posting** · posting direct-to-Highlights without going through Stories · resharing public stories · co-posting (in development) · 30 AI story effects (Jul 2026) · Generate Background · voice-over (test, Apr 2026).

5.7 Live & Broadcast channels

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Live</b></p> <ul style="list-style-type: none"><li>• <b>Gate:</b> public account, <b>≥1,000 followers</b> (since 1 Aug 2025)</li><li>• <b>Duration:</b> up to 4 h creator accounts, 1 h business accounts</li><li>• <b>Scheduling:</b> 1 hour to 90 days ahead; notifies followers at 24 h and 15 min</li><li>• Up to <b>3 guests</b> · comment pinning · <b>Practice mode</b> (invisible rehearsal) · moderator controls · fundraisers</li><li>• Saves to <b>Live Archive for 30 days</b></li></ul> | <p><b>Broadcast channels</b></p> <ul style="list-style-type: none"><li>• <b>Path:</b> DMs → pencil → "Create broadcast channel" → name → audience (all followers or a selected group) → launch</li><li>• <b>Gate:</b> Professional (Creator or Business) account. A 10,000-follower requirement is commonly cited but not universally confirmed <span>P</span></li><li>• Members react with emoji, vote in polls, see nested replies — <b>but cannot reply inline</b>. One-to-many by design.</li><li>• No API, no third-party scheduling</li></ul> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

5.8 The share screen — every field

| FIELD                  | DETAIL                                                                                                                                                                                                                                    |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Caption                | <b>2,200 character max</b> ; folds at ~125 characters behind "...more". Meta AI can draft or rewrite. Meta Verified subscribers can place <b>clickable links in captions</b> (Mar 2026).                                                  |
| Add music              | Song picker with a "Change audio" button (May 2026); trending songs surfaced. <b>Audio is replaceable post-publish</b> (Jul 2026) without losing engagement.                                                                              |
| Tag people             | Up to <b>20</b> accounts per post. Contains the "Invite collaborator" entry.                                                                                                                                                              |
| Add collaborator       | Up to 5 collaborators → the post appears on up to <b>6 profiles</b> and pools their audiences. Feed photos, carousels and Reels only — <b>not Stories</b> . Invitee must accept. Can be added after publishing via ⋮ → Edit → Tag people. |
| Add location           | Geotag; feeds the Instagram Map                                                                                                                                                                                                           |
| Add topics             | Up to <b>3</b> industry topic tags (Reels)                                                                                                                                                                                                |
| Cover                  | Reels only — frame scrubber or upload                                                                                                                                                                                                     |
| Audience               | Public / Close Friends (Stories) / Trial (Reels)                                                                                                                                                                                          |
| Trial toggle           | Reels only — see below                                                                                                                                                                                                                    |
| Also share to Facebook | Cross-post toggle. Threads too; Threads likes now surface back into Instagram.                                                                                                                                                            |

Advanced settings — the accordion everyone forgets to build

1. **Hide like and view counts** on this post
2. **Turn off commenting** (reversible later)
3. **Write alt text** — editable post-publish; Instagram auto-generates it via computer vision if left blank
4. **Upload at highest quality**
5. **Share to Facebook** (auto cross-post for synced accounts)
6. **Branded content** → paid partnership label + tag business partner; now surfaces a Creator Marketplace section (Apr 2026)
7. **Add reminder** — event countdown attached to the post
8. Reel-specific: **Recommend on Facebook** · **Allow remixing** · **Allow using your audio** · **Auto-generated captions**

**Trial Reels — the most interesting mechanic in the whole composer**

A toggle on the share screen. The Reel is **withheld from your followers** and served **only to non-followers** in waves for **72 hours**. If it clears an internal engagement threshold it **auto-publishes to followers and the grid**; otherwise it stays quietly in the Trials tab and nobody who follows you ever knows it existed.

Requires a professional account with ≥1,000 followers (opened to all qualifying accounts early 2026). No cap on concurrent trials; 30-day window; schedulable.

**Why it matters to you:** it removes the social cost of experimenting. That is a product mechanic, not a feature — and it is cheap to implement relative to how much creator behaviour it changes.

5.9 The web create flow — where it differs

Steps 1–2 below were walked live on 18 Aug 2026 and are measured, not sourced.  
The flow was stopped at the file-picker — no content was uploaded and nothing was published.

- 1

Entry — "Create" at position 7 of the left rail

VERIFIED

Icon labelled `New post` in the DOM, **Create** when the rail is expanded. Sits directly below Notifications, mid-rail — *not* at the bottom as commonly documented. Opens a popover.

**Popover contents, measured:** `Post · Live video · Ad`. ("Ad" appears on professional accounts.) **There is no Story or Reel entry in the web Create menu** — a vertical video becomes a Reel via the disambiguation prompt in the next step, not by choosing it here.

This resolves the contested "can you go Live from web?" question: yes.
- 2

"Create new post" — upload dialog

VERIFIED

Dialog measures `547 × 590 px`. Centred illustration of a photo + video card pair, headline **"Drag photos and videos here"**, and a **"Select from computer"** button — `background: rgb(74,93,249) , border-radius: 8px , font: 14px/600`.

The file input's own accept attribute, read from the DOM:

```
accept="image/avif, image/jpeg, image/png, image/heic, image/heif, video/mp4, video/quicktime" multiple="true"
```

**AVIF is accepted. GIF and BMP are not.** Only two video containers: MP4 and QuickTime MOV. This supersedes every published format list.

**Interrupt:** if the file is a vertical 9:16 video, Instagram asks **"Create post" vs "Create reel"** before continuing.
- 3

Crop

Bottom-left toolbar, three icon buttons:

  - Crop icon** → aspect menu: **Original · 1:1 · 4:5 · 16:9** (*note: web offers 16:9 where mobile offers 1.91:1*)
  - Zoom icon** → slider; drag media to reposition
  - Gallery icon** → thumbnail strip with "+" to add files, drag to reorder, hover-X to remove

Header: [`<`] "Crop" [`Next`]
- 4

Edit — two tabs

**Filters:** grid of live-preview thumbnails with an intensity slider on the selected one. Web ships a *subset* of mobile's library — roughly Original, Aden, Clarendon, Crema, Gingham, Juno, Lark, Ludwig, Moon, Perpetua, Reyes, Slumber `U`. Mobile-exclusive filters include Mayfair, Lo-Fi and X-Pro II.

**Adjustments:** brightness, contrast, saturation, fade confirmed; temperature and vignette likely `P`. That is **six sliders on web versus thirteen tools on mobile**.

Video posts skip filters entirely and instead offer **cover-frame selection from a filmstrip**.
- 5

Share — two-pane: media preview left, form right

- Avatar + username header
  - "Write a caption..." textarea, 2,200 char counter, emoji picker
  - Add location · Tag people (click the image to place a tag) · Add collaborators
  - Accessibility** accordion → alt text per image
  - Advanced settings** accordion → hide like/view counts · turn off commenting · cross-post to Facebook/Threads
  - "Share" button, top-right of the modal

| WEB CAPABILITY                      | SUPPORTED?      | NOTES                                                                                                                                                  |
|-------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Photo post                          | YES             | Unanimous across sources                                                                                                                               |
| Video post (feed)                   | YES             |                                                                                                                                                        |
| Carousel                            | YES — 20 SLIDES | Raised from 10 in 2025. One stale source still says 10.                                                                                                |
| Reels                               | YES, LIKELY     | 4 sources say yes, 2 say no. Upload a 9:16 video → "Create reel" → auto cover from frames → caption → location → Share.                                |
| Reels — custom cover upload         | NO              | Frame selection only                                                                                                                                   |
| Reels — audio library               | NO              | The biggest single web gap                                                                                                                             |
| Stories                             | NO              | Requires device emulation, Business Suite, or a third-party scheduler <code>P</code>                                                                   |
| Live video                          | YES — CONFIRMED | <b>Resolved by live inspection:</b> "Live video" is a listed item in the web Create popover. Still subject to the ≥1,000-follower public-account gate. |
| Ad                                  | YES             | Third item in the web Create popover on professional accounts — undocumented in every walkthrough consulted                                            |
| Scheduling / drafts                 | NO              | Instagram's native scheduler is mobile-only and covers posts, carousels and Reels — <b>never Stories</b>                                               |
| AR effects, stickers, text overlays | NO              | Must be pre-edited externally                                                                                                                          |
| Shopping / product tagging          | NO              |                                                                                                                                                        |

5.10 Edit tool catalogue — all 13, with ranges

| # | TOOL       | RANGE           | BEHAVIOUR                                                                               |
|---|------------|-----------------|-----------------------------------------------------------------------------------------|
| 1 | Adjust     | ±45° straighten | 90° rotate · straighten · vertical and horizontal perspective/keystone correction       |
| 2 | Brightness | −100 ... +100   | Global exposure                                                                         |
| 3 | Contrast   | −100 ... +100   | Tonal separation                                                                        |
| 4 | Structure  | −100 ... +100   | Local detail / clarity — midtone micro-contrast. Instagram's signature look lives here. |

| #  | TOOL       | RANGE                | BEHAVIOUR                                                                                                                                     |
|----|------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 5  | Warmth     | −100 ... +100        | Colour temperature; left blue, right orange                                                                                                   |
| 6  | Saturation | −100 ... +100        | Colour intensity                                                                                                                              |
| 7  | Color      | 0 ... 100 per swatch | Tint applied <b>separately to Shadows and Highlights</b> ; 8 colour swatches each, with intensity. The most sophisticated control in the set. |
| 8  | Fade       | 0 ... 100            | Lifts blacks / washes out                                                                                                                     |
| 9  | Highlights | −100 ... +100        | Recover or lift highlights only                                                                                                               |
| 10 | Shadows    | −100 ... +100        | Crush or lift shadows only                                                                                                                    |
| 11 | Vignette   | 0 ... 100            | Darkens edges                                                                                                                                 |
| 12 | Tilt Shift | on/off + size        | Selective blur — <b>Radial</b> or <b>Linear</b> mode, drag and pinch to size                                                                  |
| 13 | Sharpen    | 0 ... 100            | Fine-detail acutance                                                                                                                          |

**UNVERIFIED** Instagram does not publish numeric slider ranges. The −100...+100 (bipolar, centre-detented) / 0...100 (unipolar, left-anchored) convention above is the widely-documented behaviour. Treat it as the de-facto model.

### 5.11 Filter catalogue

The classic set still shipping:

Clarendon · Juno · Gingham · Lark · Valencia · Ludwig · Aden · Moon · Willow · Amaro · Rise · Crema · Perpetua · Mayfair · Hudson · Stinson · X-Pro II · Lo-Fi · Kelvin

Extended catalogue also documents: Nashville, Reyes, Slumber, Inkwell, Toaster, Brannan, Earlybird, Sutro, Walden, 1977, Sierra, Skyline, Dogpatch, Vesper; a family of city filters (Paris, Los Angeles, Oslo, Melbourne, Jakarta, Abu Dhabi, Buenos Aires, New York, Jaipur, Cairo, Tokyo, Rio de Janeiro); and the "modern" families **Simple / Simple Warm / Simple Cool, Fade / Fade Warm / Fade Cool, Boost / Boost Warm / Boost Cool**, plus Grainy, Gritty, Halo, Color Leak, Soft Light, Zoom Blur, Handheld, Moire, Lo-Res, Wavy, Wide Angle.

A Sept 2025 iOS test added Fade, Grainy, Graphite, Gritty, Handheld, Hyper, Lo-res, Midnight, Simple, Soft Light and Zoom Blur to the *Stories* camera specifically.

**What the filter list actually tells you**

Roughly 60 filters exist, but the tray is **user-reorderable and user-hideable**. Instagram solved filter bloat with personalisation instead of curation. If you ship 60 looks, ship the "Manage" screen on day one — otherwise the tail is dead weight that slows down the single most-used screen in your app.



# Media technical specification

Everything your transcoder, validator and upload client need to agree on. Most of this is published by Meta and is therefore solid.

## 6.1 Images

| SPEC                          | VALUE                                       | NOTES                                                                                                                                                             |
|-------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Formats accepted (web upload) | AVIF · JPEG · PNG · HEIC · HEIF · MP4 · MOV | <div>READ FROM THE LIVE FILE INPUT, 18 AUG 2026.</div> GIF and BMP are NOT accepted, contrary to most published lists. Non-JPEG images are converted server-side. |
| Formats accepted (mobile app) | Broader — adds BMP and non-animated GIF     | Per published guidance; not independently verified                                                                                                                |
| Formats accepted (Graph API)  | JPEG only                                   | Catch this in validation, not at upload                                                                                                                           |
| Max file size                 | 8 MB (API) / ~30 MB (app)                   |                                                                                                                                                                   |
| Min width                     | 320 px                                      |                                                                                                                                                                   |
| Max width                     | 1440 px                                     | Instagram downscales anything above this                                                                                                                          |
| Feed aspect range             | 1.91:1 → 4:5                                | 9:16 also accepted                                                                                                                                                |
| Square                        | 1080 × 1080                                 |                                                                                                                                                                   |
| Portrait (recommended)        | 1080 × 1350                                 | 4:5 — most feed real estate                                                                                                                                       |
| Landscape                     | 1080 × 566                                  | 1.91:1                                                                                                                                                            |
| Profile grid thumbnail        | 3:4 crop                                    | Applied to everything                                                                                                                                             |
| Profile photo                 | 320 × 320                                   | Circular mask                                                                                                                                                     |
| Stories / Reels image         | 1080 × 1920                                 | 9:16; safe area ≈250 px clear top and bottom                                                                                                                      |
| Colour space                  | sRGB                                        | Converted automatically                                                                                                                                           |

### Compression behaviour — build your pipeline around this

Instagram re-encodes every upload as JPEG at roughly 70–75% quality, downscales the long edge to the 1080/1440 ceiling, and strips all EXIF metadata — GPS, camera and lens, shutter, ISO, aperture. Uploading pre-sized at exactly 1080 px wide avoids the downscale pass but not the re-encode.

User-facing control: Settings and privacy → Data usage and media quality → disable Data Saver, enable Upload at highest quality. Off by default on both platforms.

#### The EXIF strip is a privacy feature, not an optimisation

Stripping GPS from every uploaded photo is the single highest-leverage privacy decision in Instagram's media pipeline, and it happens silently server-side. If you build a photo app and skip this, you will eventually ship a story about a user's home address being extractable from their posts. Strip metadata by default, at ingest, before the file ever lands in durable storage.

## 6.2 Video, per surface

|                    | REELS                                                                           | STORIES                          | CAROUSEL VIDEO                   | LIVE                 |
|--------------------|---------------------------------------------------------------------------------|----------------------------------|----------------------------------|----------------------|
| Recommended res    | 1080 × 1920                                                                     | 1080 × 1920                      | 1080×1080 / 1080×1350 / 1080×566 | 1080 × 1920          |
| Min res            | 540 × 960                                                                       | —                                | —                                | 720p                 |
| Max height / width | 3600 px                                                                         | 3600 × 1920                      | —                                | —                    |
| Aspect ratio       | 9:16 rec.; 1.91:1 → 9:16 accepted                                               | 9:16 rec.; 0.1:1 → 10:1 accepted | 1:1 / 4:5 / 1.91:1               | 9:16                 |
| Duration min       | 3 s                                                                             | 3 s (API) / ~1 s (app)           | —                                | —                    |
| Duration max       | 15 min (API) · 20 min (in-app camera) · >3 min not recommended to non-followers | 60 s per segment                 | 60 s per slide                   | 4 h                  |
| Max file size      | 1 GB (API) / ~4 GB app                                                          | 100 MB (API) / ~30 MB app        | 4 GB                             | —                    |
| Container          | MOV or MP4 (MPEG-4 Part 14), no edit lists, moov atom at front                  |                                  | MP4 / MOV                        | —                    |
| Video codec        | HEVC or H.264 · progressive scan · closed GOP · 4:2:0 chroma                    |                                  | H.264                            | H.264                |
| Audio codec        | AAC · ≤48 kHz · 1–2 channels                                                    |                                  | AAC                              | AAC                  |
| Frame rate         | 23–60 fps                                                                       | 23–60 fps                        | ≥30 fps                          | ≥30 fps              |
| Video bitrate      | VBR, 25 Mbps max                                                                | 25 Mbps max                      | ~3,500 kbps                      | 3,500–5,000 adaptive |
| Audio bitrate      | 128 kbps                                                                        | 128 kbps                         | —                                | —                    |
| Thumbnail          | JPEG, ≤8 MB, 9:16                                                               | —                                | —                                | —                    |

#### "Max Reel length" is three different numbers and they are all correct

**3 minutes** — the recommendation-eligibility cliff. Longer than this and Instagram stops serving it to non-followers. This is the number that actually matters commercially.  
**15 minutes** — the Graph API upload ceiling. This is the number your API integration will hit.  
**20 minutes** — the in-app camera recording ceiling, raised Nov 2025. This is the number a user will quote at you.  
Legacy 15/30/60/90 s presets also persist in the length picker. Spec all four separately or your validation will be wrong for someone.

6.3 Hard limits — the full table

| LIMIT                          | VALUE           | NOTES                                                                                    |
|--------------------------------|-----------------|------------------------------------------------------------------------------------------|
| Caption                        | 2,200 chars     | All post types                                                                           |
| Caption fold                   | ~125 chars      | Before "...more"                                                                         |
| Hashtags per post              | 5               | Reduced from 30, late 2025 → 2026. Some outlets observed 3 in test cells. <span>P</span> |
| Hashtag following              | Removed         | 13 Dec 2024                                                                              |
| Tagged accounts per post       | 20              |                                                                                          |
| Collaborators                  | 5               | 6 profiles total                                                                         |
| Carousel items                 | 20 app / 10 API |                                                                                          |
| Bio                            | 150 chars       | 1 clickable link; Link Pages for Meta Verified (2026)                                    |
| Username                       | 30 chars        | a–z, 0–9, ., _                                                                           |
| Display name                   | 30 chars        |                                                                                          |
| Comment                        | 1,000 chars     |                                                                                          |
| DM message                     | 1,000 chars     |                                                                                          |
| Notes                          | 60 chars        | 24 h (3-day variant exists); mutuals / Close Friends                                     |
| Story text sticker             | ~250 chars      |                                                                                          |
| Stories per 24 h               | 100             |                                                                                          |
| Accounts you can follow        | 7,500           | Hard ceiling                                                                             |
| Group DM size                  | 250             |                                                                                          |
| Pinned chats / pinned messages | 3 / 3           |                                                                                          |
| Voice note                     | 5 min           |                                                                                          |
| API posting rate               | 50 posts / 24 h |                                                                                          |

Behavioural anti-spam limits (unofficial, observed): new accounts ~100 follows, ~250 likes, 30–50 comments, 20–50 DMs per day. Established accounts 300–500 follows, ~1,000 likes, 100–150 comments, 50–150 DMs per day. Worth replicating in spirit — rate limits on *social actions*, not just API calls, are what keep a new network from being immediately overrun.

6.4 Music

| SURFACE          | CLIP LENGTH                                                                                                                             |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Stories          | 15 s default, extendable to segment length; <b>raised to 45 s</b> Oct 2025. Music-sticker clip reported at 20 s Apr 2026 <span>P</span> |
| Reels            | Up to <b>60 s</b> for most library tracks; <b>imported audio uncapped</b>                                                               |
| Feed photo posts | Supported via "Add music" (2025+); auto-added when a post is shared to Story                                                            |

**Licensing reality: Business accounts have restricted access to the commercial music library.** The workaround is a Creator account or importing licensed audio. Post-publish **audio replacement** on feed posts and carousels shipped 21 July 2026 (post → ⋯ → Edit), preserving engagement.

**Music licensing is the hardest non-obvious problem in this entire document**

Instagram's music library exists because Meta signed direct deals with every major label and thousands of publishers. You cannot replicate that at launch, and using unlicensed commercial music will get your app removed from the App Store and sued — in that order. **Realistic paths:** license a production-music catalogue (Epidemic Sound, Artlist, Musicbed all have B2B API tiers), let users import their own audio and put the liability on them via ToS plus Content ID-style fingerprint matching, or ship with no music at launch. Budget this as a business-development problem with a legal cost, not an engineering ticket.

# Build notes for your own app

You asked for a full design spec, not a build guide — so this part is deliberately short. It exists to stop three specific expensive mistakes.

## 7.1 A reality check on scope

Everything catalogued in Parts 1–6 represents roughly **110 mobile destinations, 30+ web routes, ~15 core components, 6 distinct upload pipelines, ~60 filters, 13 edit tools and ~30 sticker types** — built over fourteen years by a team that grew into the thousands, on top of Meta's existing CDN, transcoding, ranking and moderation infrastructure.

The useful question is not "how do I build all of that." It is **"which 8% of it is the product?"** Instagram's own 2010 v1 was:

**Take a photo → apply one of 11 filters → post it to a square grid → follow people → see their photos in a reverse-chronological feed → like → comment.**

No video. No Stories. No DMs. No Explore. No algorithm. No web app. It hit one million users in two months.

## 7.2 Suggested build order

| PHASE | SCOPE                                                                                                               | WHY THIS ORDER                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1     | Auth · profile · single photo upload with crop · 3-column grid · follow graph · chronological feed · like · comment | This is a complete social product. If it isn't compelling here, no amount of Reels will save it.                             |
| 2     | Carousel · filters & edit sliders · alt text · post detail modal · notifications                                    | Carousel is the cheapest engagement multiplier in the whole feature set — it is just an array on an existing object.         |
| 3     | DMs · Stories with a 24 h TTL · story ring states                                                                   | Stories are a <i>different</i> media object with an expiry job, not a variant of a post. Model them separately from day one. |
| 4     | Short video / Reels · transcoding pipeline · vertical player                                                        | The expensive one. Video is where your infrastructure bill becomes real.                                                     |
| 5     | Explore · ranking · search · moderation tooling                                                                     | Only meaningful once you have enough content that chronological stops working.                                               |

## 7.3 Three data-model decisions worth stealing

**1. One media object, many surfaces**

Instagram collapsed feed video into Reels in 2022 and the product got simpler. Model `Media` once — with a `surface` enum (feed, reel, story) and a `ttl` — rather than `Post` , `Video` , `Reel` and `Story` as four parallel hierarchies.

**2. Carousel is an ordered array on the media object, not a separate type**

And **the aspect ratio lives on the parent, not the child** — that is exactly why slide 1 sets the ratio for everything. That constraint is a schema decision surfacing as a UX rule.

**3. Derivatives are generated at ingest, never on read**

Instagram produces the 1080-wide feed render, the 3:4 grid crop, the 320×320 avatar and the story render at upload time. Every read is a CDN hit on a pre-baked file. Resist on-the-fly resizing — it is the classic mistake that makes a photo app fall over at exactly the moment it starts working.

**Media pipeline: the four things that will bite you**

**Strip EXIF at ingest** (privacy, non-negotiable) · **put the moov atom at the front of every MP4** or video won't start streaming until fully downloaded · **generate a poster frame server-side**, never rely on the client · **cap and validate before transcode**, because a 4 GB upload that fails validation after transcoding has already cost you the money.

# What you may and may not copy

Building "an app like Instagram" is entirely legal. Building "Instagram" is not. The line is clearer than most people assume — and this document sits on both sides of it, so read this before handing it to a designer.

**Not legal advice**

I'm not a lawyer and this isn't legal advice. It's a practical summary of where the well-established boundaries sit, so you know which questions to take to actual counsel before launch. Get a real IP review before you ship — it is cheap compared to a rebrand after.

## 8.1 The three separate things people conflate

| TYPE                  | WHAT IT PROTECTS                                                                                                                                                     | YOUR POSITION                                                                                                                                                                                                   |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Functionality         | What the software <i>does</i> — a photo feed, double-tap to like, stories that expire in 24 hours, a 3-column grid, filters, carousels, an algorithmic Explore page. | <b>FREE TO COPY.</b> Functional concepts are generally not protectable by copyright. Every one of these has been implemented by a dozen apps. Snapchat invented Stories; Instagram copied it wholesale and won. |
| Trade dress           | The distinctive <i>look and feel</i> that consumers use to identify the source — the gradient, the icon, the specific colour-plus-layout signature.                  | <b>RISKY ZONE.</b> Copying enough of the visual identity that a normal user could confuse your app for Instagram is actionable. This is where careless clones die.                                              |
| Trademark & copyright | The name "Instagram", the wordmark, the camera glyph, the app icon, Instagram Sans, their actual code, their actual icon SVGs.                                       | <b>OFF LIMITS, ABSOLUTELY.</b> No amount of "inspired by" helps here.                                                                                                                                           |

## 8.2 Practical rules for using this document

**Safe to take directly**

- Every **flow** in Part 5 — step sequences aren't protectable
- Every **technical spec** in Part 6 — codecs, ratios, limits are facts
- The **token architecture** in Part 3.1 — RGB triplets, two layers, class-based themes
- The **information architecture** — tabs, routes, screen inventory
- **Layout proportions** — a 3-column grid and a ~470px feed column are functional choices
- The **component patterns** in Part 4 — stacked options sheets, slide-over panels
- Interaction conventions — double-tap to like, swipe between stories, pull to refresh

**Do not take**

- The **5-stop brand gradient** — this is the single most identifying asset Instagram has, more than the wordmark
- The **camera glyph** or anything close to it
- The **name**, or anything with "insta" or "gram" in it — Meta's brand guidelines explicitly police both
- **Instagram Sans** — not licensable to you at all
- Their **actual icon SVGs** or CSS bundle
- **Filter names** — "Clarendon", "Juno", "X-Pro II" are branded looks. Copy the *effects*, rename them.
- Scraped **user content or data** from Instagram to seed your app

**The practical test**

Put your app's home screen next to Instagram's on a phone. Could a distracted person tap the wrong one? If yes, change your colour signature and your icon — not your feature list. Differentiation that matters lives in *identity*, not in *functionality*. TikTok, Pinterest, VSCO and BeReal all share most of Instagram's mechanics and nobody confuses any of them for it.

**One more, on data**

Part 0 of the original conversation covered scraping. To restate it here because it matters at build time: **do not seed your app with content scraped from Instagram**. Unofficial private-API libraries violate Meta's terms, reliably get accounts banned, and importing other people's photos into your product creates copyright exposure per photo. Seed with your own content, invited creators, or a licensed stock corpus.

# Open conflicts & sources

## A.0 Resolved by the live pass — no longer open

| ITEM                             | RESOLUTION                                                                             |
|----------------------------------|----------------------------------------------------------------------------------------|
| Primary button colour            | #4A5DF9 indigo. Not #0095F6 , not black/white. The purple-accent reports were correct. |
| Live video from web              | Yes — listed in the Create popover                                                     |
| Web vs app navigation divergence | No divergence — the web rail runs the same Reels-2nd / Messages-3rd order              |
| Grid tile ratio                  | 3:4 confirmed — 0.750 measured                                                         |
| Grid gutter                      | 1px, not ~4px                                                                          |
| Content container width          | 680px on profile/Explore; 1457px outer shell. 935px is legacy.                         |
| Collapsed rail width             | ~72px confirmed (48px item, 12px inset)                                                |
| Accepted upload formats (web)    | AVIF, JPEG, PNG, HEIC, HEIF, MP4, MOV. No GIF, no BMP.                                 |
| Button radius / type scale       | 8px radius, 14px/600 confirmed on the live CTA                                         |
| Missing <nav> landmark           | Confirmed — still no navigation landmark in the DOM                                    |

## A.1 Still unresolved

| ITEM                                                                                                                                                                                                                                                                     | STATUS                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Create "+" placement on mobile                                                                                                                                                                                                                                           | Top-left (3 sources) vs top-right (2 sources). Likely a live A/B or a mid-rollout change. <i>Not resolvable from the web app.</i>                                  |
| Max Reel duration                                                                                                                                                                                                                                                        | 20 min (feature page + in-app camera) / 15 min (Graph API) / 3 min (recommendation cliff). All three real.                                                         |
| Hashtag cap                                                                                                                                                                                                                                                              | 5 (better sourced) vs 3 (test cells).                                                                                                                              |
| Story audio length                                                                                                                                                                                                                                                       | 45 s (Oct 2025) vs 20 s music sticker (Apr 2026). Different limits or a rollback.                                                                                  |
| Story file caps                                                                                                                                                                                                                                                          | API: 8 MB image / 100 MB video. App-side guides: 30 MB / 4 GB. Use API numbers for anything API-adjacent.                                                          |
| Broadcast channel follower gate                                                                                                                                                                                                                                          | 10,000 vs no stated follower requirement.                                                                                                                          |
| Stories from web                                                                                                                                                                                                                                                         | 3 say no, 1 says yes. Lean no.                                                                                                                                     |
| Reels from web                                                                                                                                                                                                                                                           | 4 say yes, 2 say no. Lean yes.                                                                                                                                     |
| Live from web                                                                                                                                                                                                                                                            | 2 say yes, 1 says no. Likely account-gated.                                                                                                                        |
| Profile grid reordering                                                                                                                                                                                                                                                  | Shipping vs removed. Assume unavailable.                                                                                                                           |
| Desktop profile avatar                                                                                                                                                                                                                                                   | Measured 120 × 120 at an 879px viewport — likely scales to 150 at wider widths. Neither 150 nor 180 confirmed at desktop breakpoint.                               |
| Feed post card width                                                                                                                                                                                                                                                     | Still open. The verification account followed zero accounts, so the home feed rendered empty and no <article> existed to measure. ~470px remains a reconstruction. |
| Expanded rail width                                                                                                                                                                                                                                                      | ~244px unconfirmed — the session ran at an 879px viewport where the rail is permanently collapsed.                                                                 |
| Still unverified: feed post card and right-rail widths · story tray geometry · post-detail modal dimensions · the exact web filter list and full Adjustments list · the mobile-web bottom tab bar composition · every mobile-app measurement (no device access).         |                                                                                                                                                                    |
| How to close the remaining gaps in ten minutes: open instagram.com on an account that follows at least one person, at a viewport ≥1280px, and run the same measurement pass on document.querySelector('article') , the story tray container, and the post-detail dialog. |                                                                                                                                                                    |

## A.2 Sources

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Instagram / Meta official</p> <p>about.instagram.com — Brand: Gradient · Reels feature page · Favorites and Following announcement · New stickers for Stories (May 2024) · sitemap</p> <p>meta.com/brand/resources/instagram — Brand Resource Center</p> <p>spark.meta.com — Meta Spark shutdown announcement</p> <p>help.instagram.com — Your Activity; advanced settings</p> <p>Meta Graph API media guidelines (via Ayrshare mirror)</p> <p>Technical extraction</p> <p>projectwallace.com — A deep dive into the CSS of Instagram.com (breakpoints, custom-property count, CSS budget)</p> <p>gist.github.com/etekweb — Instagram dark-mode CSS variable dump</p> <p>github.com/vednoc/dark-instagram — userstyle source (token roles, 935px)</p> | <p>shadcn.io/design/threads — Threads design system reverse-engineering</p> <p>dustinwhisman.com — accessibility audit of Instagram</p> <p>Product &amp; rollout reporting</p> <p>TechCrunch · Engadget · Social Media Today · PetaPixel · Tubefilter · PiunikaWeb · NetInfluencer · BuzzInContent · Lindsey Gamble</p> <p>Practitioner documentation</p> <p>Storrito · Inrō · Metricool · SocialBee · SocialRails · SocialPilot · Social Champ · Buffer · Hootsuite · Oktopost · Kapwing · Brandwatch · brandID · Taplink · PostEverywhere · HopperHQ · SocialzAI · Keesy · Sendible · Sendcove · NapoleonCat · ClickAnalytic · Minter.io · Adobe Express · iPhone Photography School · Bustle · Viallo · Swarmify · CircleCropImage · PostFast · GIJN · Outfame · LouiseM · color-hex · ColorKit · USBrandColors · BrandPalettes · FontsArena · Vaizle</p> |
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This document is independent competitive and interface research prepared for product development purposes. It reproduces no Instagram source code, assets or user content.

Compiled 18 August 2026 · No live logged-in session was available at time of authoring · Confidence tags apply throughout.